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DEVICES, SYSTEMS, AND METHODS FOR MAINTAINING A STERILE SURGICAL FIELD

Abstract

A medical system comprises an arm; a medical device holder coupled or mountable to the arm; a drape configured to cover the medical device holder and at least a portion of the arm in a coupled or mounted state of the medical device holder to the arm; and a cable guide. The cable guide comprises a base portion coupled or configured to be coupled to a complementary receptacle of the arm; and two opposing guidepost arms extending from the base portion and defining an opening configured to receive and route a transmission line. In a coupled state of the base portion to the receptacle, the guidepost arms are positioned on a first side of the drape and the arm is positioned on a second side of the drape, the first side being opposite from the second side.

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Background/Summary

CROSS-RELATED APPLICATIONS [0001] This application is a divisional application of U.S. application Ser. No. 17/435,100, filed Aug. 31, 2021, which is a U.S. national stage application under 35 U.S.C. § 371(c) of International Application No. PCT/US2020/021139, filed Mar. 5, 2020, which claims priority to and the benefit of the filing date of U.S. Provisional Patent Application 62/813,973, filed Mar. 5, 2019 (now expired), the entire contents of each of which are incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] Aspects of the present disclosure relate to devices, systems, and methods for maintaining sterility of a surgical field while utilizing various instruments and devices that may be nonsterile or make contact with equipment in a nonsterile field.

INTRODUCTION

[0003] Minimally invasive surgery seeks to minimize patient trauma by introducing therapeutic, diagnostic, and imaging instruments through small incisions or natural orifices. One group of such instruments are manually operated. Another group of such instruments are teleoperated by using a computer-assisted surgical system (a telesurgical system), in which a surgeon operates an input control unit to remotely control one or more instruments operated by a manipulator unit to which the one or more instruments are coupled.

[0004] Whether operated manually or via a telesurgical system, many instruments are coupled to auxiliary equipment that support the instruments' clinical purpose. For example, electrosurgical instruments are coupled to electrosurgical generator units (ESUs) that provide mono- and bipolar energy to the instrument as required. In addition, other auxiliary equipment and associated medical devices may be used to support a surgical procedure. For example, suction/irrigation devices that may be used during various surgeries require operable coupling to corresponding irrigation fluid and vacuum sources. Similarly, endoscope instruments and other imaging devices require supporting imaging and illumination equipment. Thus, various auxiliary function equipment is typically used during minimally invasive surgery to support medical devices used at a remote surgical site in performing functions such as insufflation supply, cautery smoke evacuation, ultrasonic energy generation, etc. As used herein, medical devices include both instruments used for manipulating tissue and sensing an environment (e.g., imaging, pressure, oxygen etc.) at a remote surgical site, as well as devices used to supply suction, irrigation, light, and other types of flux proximate a remote surgical site.

[0005] As a result of the many auxiliary function units that are used to support modern surgery, and the various connections (e.g., via transmission lines) between the auxiliary function equipment and the corresponding medical devices, the operating room becomes a complex and crowded environment during surgery. For example, maintaining a sterile field around a patient during surgery can pose challenges. In particular, it can be challenging to maintain sterility in applications

that involve use of a variety of medical devices to perform a surgical procedure that may come into and out of use and/or require connection via transmission lines (e.g., electrical and/or data cables) to equipment that is typically not sterile and is maintained outside of the sterile field. For example, in various surgical applications, including various minimally-invasive surgical applications, medical devices, such as electrosurgical instruments, insufflation/evacuation tubing, imaging instruments, and the like are connected to auxiliary function equipment that supply and control the operation of those medical devices. Such auxiliary function equipment may be in a portion of the operating theatre that is not part of the sterile surgical field, and is therefore generally not sterilized. Providing robust and continued connection of medical devices in the sterile field to such equipment outside the sterile field is thus needed to ensure surgical procedures can occur efficiently and sterile conditions maintained.

[0006] A need exists to facilitate connection of the various medical devices that are used during a surgical procedure and are in the sterile field to equipment outside the sterile field without risking a breach of the sterile field or medical devices used in the sterile field. Further, a need exists to provide a convenient and accessible place to hold surgical instruments and other medical devices while not in use, but that maintains their sterility. In addition, a need exists to provide medical devices with a relatively large reach into, and range of movement within, the sterile field while still allowing them to be connected via transmission lines to equipment outside of the sterile field. A need also exists to manage transmission lines during surgical procedures that may have multiple instruments and multiple transmission lines running from the sterile field to auxiliary equipment outside the sterile field.

SUMMARY

[0007] Exemplary embodiments of the present disclosure may demonstrate one or more of the above-mentioned desirable features. Other features and/or advantages may become apparent from the description that follows.

[0008] Various embodiments include a sterile drape assembly comprising a drape and a cable guide coupled or configured to be coupled to the drape. The drape may comprise a first portion configured to cover a medical device holder and a second portion configured to cover a portion of an arm that supports the medical device holder. The cable guide may comprise two opposing guidepost arms extending in a first direction and defining a first opening configured to receive and route a first transmission line along a first path. The cable guide may also comprise a side arm extending laterally from one of the guidepost arms, the side arm defining a second opening spaced from the first opening and is configured to receive and route a second transmission line along a second path spaced from the first path.

[0009] Various embodiments also include a method comprising arranging a drape over an arm supporting a medical device holder and attaching a cable guide to the arm by inserting a base portion of the cable guide into a complementary receptacle in the arm such that a portion of the cable guide is on a first side of the drape and the receptacle is on a second side of the drape opposite the first side of the drape.

[0010] Various embodiments also include a medical system, comprising an arm, a medical device holder coupled or mountable to the arm, a drape configured to cover the medical device holder and at least a portion of the arm in a coupled or mounted state of the medical device holder to the arm, and a cable guide. The cable guide may comprise a base portion coupled or configured to be coupled to a complementary receptacle of the arm. The cable guide may further comprise two opposing guidepost arms extending from the base portion and defining an opening configured to receive and route a transmission line. In a coupled state of the base portion to the receptacle, the guidepost arms are positioned on a first side of the drape and the arm is positioned on a second side of the drape, with the first side being opposite from the second side.

[0011] For a fuller understanding of the nature and advantages of the present teachings, reference is

made to the following description and drawings. Other aspects, objects, and advantages of the present teachings will be apparent from the drawings and detailed description that follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The present disclosure can be understood from the following detailed description, either alone or together with the accompanying drawings. The drawings are included to provide a further understanding of the present disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate one or more exemplary embodiments of the present teachings and together with the description serve to explain certain principles and operation.

[0013] FIG. 1 shows a schematic illustration of a system as described herein.

[0014] FIGS. 2A-1 to 2J show several views of various exemplary embodiments of medical device holders in accordance with the present disclosure.

[0015] FIGS. 3A-1 to 4 show several views of various exemplary embodiments of medical device holder accessories in accordance with the present disclosure.

[0016] FIG. 5A shows a perspective view of an exemplary embodiment of an arm in an extended configuration in accordance with the present disclosure.

[0017] FIG. 5B shows a perspective view of an exemplary embodiment of the arm of FIG. 5A with a medical device holder attached in accordance with the present disclosure.

[0018] FIGS. 6A-6C show perspective views of extended, partially extended, and retracted configurations of an exemplary embodiment of an arm relative to a base in accordance with the present disclosure.

[0019] FIGS. 6D and 6E show further illustrations of various extended configurations of the arm of FIGS. 6A-6C with a medical device holder attached relative to a base in accordance with the present disclosure.

[0020] FIG. 7 shows distal portions of an exemplary embodiment of an arm in a raised configuration in accordance with the present disclosure.

[0021] FIG. 8 shows a perspective, cross-sectional view of components of an exemplary joint for an arm in accordance with the present disclosure.

[0022] FIG. 9A shows a perspective view of another exemplary embodiment of an arm in an extended configuration in accordance with the present disclosure.

[0023] FIG. 9B shows a perspective view of a linkage of the arm of FIG. 9A in accordance with the present disclosure.

[0024] FIGS. 10A-1 and 10A-2 show partial cut-away, perspective views of an exemplary embodiment of arm and subcomponents thereof in accordance with the present disclosure.

[0025] FIG. 10A-3 shows a side elevational view of an exemplary embodiment of the arm of FIGS. 10A-1 and 10A-2.

[0026] FIGS. 10B-1 and 10B-2 show views of an exemplary embodiment of an arm with another embodiment of a medical device holder in accordance with the present disclosure.

[0027] FIG. 10B-3 shows a schematic illustration of another exemplary embodiment of a medical device holder in accordance with the present disclosure.

[0028] FIGS. 11A and 11B show perspective views of exemplary embodiments of various arms, holders, and drapes in accordance with the present disclosure.

[0029] FIGS. 11C and 11D show perspective views of the drapes of FIGS. 11A and 11B when laid flat.

[0030] FIGS. 11E and 11F show further views of exemplary embodiments of arms, drapes, and medical device holders in accordance with the present disclosure.

[0031] FIGS. 12A and 12B show perspective and plan views, respectively, of an exemplary

embodiment of a combined drape and medical device holder in accordance with the present disclosure.

[0032] FIG. 12C shows a perspective new of an exemplary embodiment of a clip for retaining the drape of FIGS. 12A and 12B.

[0033] FIGS. 13A and 13B show perspective views of and exemplary embodiment of an end cap in accordance with the present disclosure.

[0034] FIGS. 14A-1 and 14A-2 show exploded, perspective and top views, respectively, of an exemplary embodiment of a cable guide in accordance with the present disclosure.

[0035] FIGS. 14B-1 to 14B-3 show exploded and perspective views of another exemplary embodiment of a cable guide in accordance with the present disclosure.

[0036] FIG. 14C shows an exploded view of an exemplary embodiment of attaching a cable guide to a drape in accordance with the present disclosure.

[0037] FIG. 14D shows perspective view of yet another embodiment of a cable guide in accordance with the present disclosure.

[0038] FIGS. 15A and 15B show perspective views of various exemplary embodiments of a combination of a base, arm, holder, and drape in accordance with the present disclosure

[0039] FIGS. 15C and 15D show detailed views of portion of the base of FIGS. 15A or 15B.

[0040] FIGS. 16A-16B show perspective views of exemplary embodiments of a medical device holder and an arm used with a stand.

[0041] FIG. 17 shows a perspective view of another exemplary embodiment of an arm and medical device holder including exemplary locking and breakaway mechanisms.

[0042] FIG. 18 shows a cross-sectional view, with portions broken away, of an exemplary embodiment of a breakaway mechanism.

[0043] FIG. 19 shows a perspective of an exemplary embodiment of a locking mechanism.

[0044] FIG. 20 shows another view of the embodiment of FIG. 19 in a folded configuration.

[0045] FIG. 21 shows a perspective view, with portions transparently shown, of another exemplary embodiment of an internal lock mechanism for a link member.

[0046] FIG. 22 shows a perspective view, with portions broken away, of the embodiment of FIG. 21.

[0047] FIG. 23 shows a perspective view, with portions broken away, of the embodiment of FIG. 21 in an unlocked configuration.

[0048] FIG. 24 shows a perspective view, with portions broken away, of an exemplary embodiment of a medical device holder locking mechanism.

[0049] FIG. 25 shows a perspective view, with portions broken away, of the embodiment of FIG. 24 in an unlocked configuration.

[0050] FIG. 26 shows a perspective reverse angle view of the configuration shown in FIG. 25.

DETAILED DESCRIPTION

[0051] The various disclosed embodiments include a medical device holder configured to securely hold one or more medical instruments. Thus, the various exemplary embodiments of medical device holders disclosed herein may prevent or reduce a risk of medical instruments rolling or otherwise falling off the holder. In this manner, damage to and potential reduction in the lifespan of medical instruments, which may be vulnerable devices with delicate components that can become damaged if they fall a holder, can be decreased. Additionally, various disclosed embodiments of medical device holders allow the holder to be easily moved in different positions and locations during a medical procedure. This can provide a more efficient workflow system, allowing users to easily move the holder in different positions depending on the specific procedure being performed at any given time frame during the medical procedure.

[0052] Various embodiments discussed herein also provide instrument and cable organization, thus also contributing to a more efficient workflow system. For example, embodiments of a medical device holder can include mechanisms to organize the instruments and cables so that a user may

easily know the location of each instrument before reaching for the instrument, thus improving the user's efficiency during the medical procedure. Additionally, such organization features may route cables on the medical device holder so that they do not interfere with the tips of the medical instruments when disposed on the holder. Furthermore, the cable organization and routing structures in accordance with various exemplary embodiments can separate monopolar and bipolar cables in order to prevent capacitive coupling between such cables.

[0053] Various disclosed embodiments also provide a medical device holder that may be sterilizable and/or covered with a sterilized drape.

[0054] Various disclosed embodiments of the medical device holder may include one or more components that assist with organization, access to, and maintaining sterility of various medical devices that may be used during a surgical procedure. Additionally, a medical device holder has a sufficient length to receive a shaft and back end housing mechanisms of various sized surgical instruments designed for use with telesurgical systems. The medical device holder may also have a width sufficient to hold two or more such surgical instruments. Medical device holders in accordance with various embodiments of the disclosure also can facilitate ease of swapping out medical devices, such as imaging instruments, during a surgical procedure by providing a support area large enough to easily and safely hold two or more medical devices at the same time.

[0055] Various disclosed embodiments according to the present disclosure include an arm configured for attachment to a base, one or more medical device holders, a monitor, etc. Such an arm may be an arm having a single link or an arm having multiple kinematic links. Additionally, the disclosed arm may be movable (both horizontally and vertically) and pivotable relative to the structure to which the arm is attached. Various disclosed embodiments provide an arm that facilitates an efficient workflow system by allowing the arm to be moved into multiple positions relative to the structure to which the arm is attached. Similarly, the arm-medical device holder connection can permit a variety of orientations of the medical device holder relative to the arm and thus within the operating theatre. In various embodiments, the arm and medical device holder can provide a wide reach and range of positions within the surgical sterile field while still allowing medical devices to remain connected to equipment outside the sterile field, with a minimal risk of a breach in the sterile field due to unintended contact with the non-sterile field.

[0056] An arm, in the various exemplary embodiments discussed herein, allows sterilized medical devices to be staged prior to surgery. For example, the arm may be attached to the medical device holder, and the medical devices may be disposed on the medical device holder such that their transmission lines are neatly routed along the arm, which can be provided with a sterile drape, before the medical procedure. Clips and/or posts on the arm help to organize and route the cables along the arm, allowing convenient and easy access for personnel outside the sterile field to access connectors on transmission lines to connect them with auxiliary function equipment outside the sterile field.

[0057] Various disclosed embodiments also include an arm that can assume different lengths. For example, in some embodiments the arm may be foldable and retractable in order to assume the different lengths. Such may also contribute to the efficient workflow system by allowing a user to easily move the arm (and thus the medical instruments disposed on the holder) to various positions during the course of a medical procedure. For example, the arm may move from a first position disposed over a patient to a second position remote from the patient but still within the sterile field. Moreover, the ability to fold or move from an extended length to a retracted length can provide for convenient stowage of the arm when not in use.

[0058] Various disclosed embodiments also include a drape configured to provide a sterile environment during a surgical procedure. The drape has a size and shape tailored for the structure over which the drape is disposed. The various embodiments include drapes comprising different materials in order to optimize the use of each portion of the drape. For example, a portion of the drape, which receives a medical instrument, may include an absorbable material.

[0059] A drape, according to the various embodiments discussed herein, include one or more troughs for storage of medical instruments, cables, etc., which may help to provide a more efficient workflow system by allowing the medical instruments, their transmission lines, and other devices that may be used during a surgical procedure to be neatly organized. Additionally, medical instruments may be conveniently and ergonomically stored during the medical procedure so that a user may easily access the instruments during the procedure.

[0060] Various disclosed embodiments also include surgical drapes comprising various transmission line routing features that can be positioned to extend along a length of draped parts to provide organization and routing of transmission lines from medical instruments on the medical device holder back to equipment outside the sterile field. Keeping such transmission lines routed and organized reduces personnel accidentally snagging or tripping on the transmission lines, which can cause damage to medical devices and/or personnel. It also reduces the risk of mixing up transmission lines associated with different medical devices. In addition, such routing features can allow for appropriate slack to be let in and out of transmission lines as a medical device holder that holds the medical devices is moved throughout the sterile field, which can in turn ensure medical device transmission lines remain connected to auxiliary equipment and the medical devices remain held on the medical device holder regardless of movement and repositioning of the medical device holder, including by virtue of repositioning of an arm to which the medical device holder is attached.

[0061] Various disclosed embodiments also include a base that may have a variety of forms and be configured to store and house the arm and/or medical device holder. Such a base can include a storage compartment sized to receive the arm with or without a medical device holder attached. In various exemplary embodiments, the base can be wheeled for easy transport to, from, and with an operating room. Further, in an exemplary embodiment, the base is an auxiliary system that holds one or more auxiliary function supply equipment that provides auxiliary functionality to medical devices used during a surgical procedure.

[0062] With reference to FIG. 1, a system **1** for holding medical equipment in accordance with various exemplary embodiments is depicted schematically. The system **1** includes a base **10** to which a medical device holder **20** is attached through a mechanical arm **30**. The medical device holder **20** may comprise a tray, for example. In some embodiments, medical device holder **20** is connected to base **10** through arm **30**. In some embodiments, the arm **30** and/or medical device holder **20** may be removably coupled to the base **10**. Similarly, the medical device holder **20** may be removably coupled to the arm **30**. In some embodiments, the system **1** may further include a sterile drape **40** that is disposed over medical device holder **20** and/or arm **30** (both shown in FIG. 1) to provide a sterile barrier. In some embodiments, the system **1** may include a sterile drape **40** can be a flat sheet that is disposed around medical device holder **20** and/or arm **30** to provide a sterile barrier, that is, the sterile drape **40** has a size and a shape so that it envelopes and surrounds the medical device holder **20** and/or arm **30** on some or all of its/their sides. In other embodiments, the sterile drape **40** may be configured as an enclosed sleeve-like object to receive the arm **30** and/or medical device holder **20**. Various exemplary embodiments of drapes, which may be used as the sterile drape **40**, are discussed further below.

[0063] Base **10** may be a cart and/or an auxiliary system that includes various auxiliary function equipment, such as for image processing, electrocautery energy generation, system central processing, and/or insufflation/evacuation. In other embodiments, base **10** may be any cart or stationary base for attachment to medical device holder **20** and arm **30**. For example, the various embodiments of the medical device holder **20** and arm **30** can be provided as part of a moveable stand, similar to a Mayo stand. Alternatively, the arm **30** and medical device holder **20** may be configured to be coupled to other components of an operating theatre, such as a patient bed or a manipulator system in a telesurgical system.

[0064] As discussed further below, in an exemplary embodiment, medical device holder **20** and arm

30 may be housed in a base **10** when not in use, which may be convenient to move the base **10**, and consequently the holder **20** and arm **30**, out of the way when not in use. In preparation for a surgical procedure, medical device holder **20** and arm **30** may be removed from a storage position within the base **10** such that medical device holder **20** is placed in proximity, including over, a patient's bed. Drape **40** may also be disposed over medical device holder **20** and arm **30** in order to provide a sterile environment. One or more medical devices, such as imaging instruments (e.g., endoscopes), surgical instruments, and/or insufflation/evacuation tubing, may be disposed over sterile drape **40** and supported by medical device holder **20**. Medical device holder **20**, arm **30**, and base **10** may allow a surgeon to easily access the medical devices held by medical device holder **20** during the surgical procedure.

[0065] The system **1** may further include a variety of features that provide for efficient management and connection of transmission lines that connect the various medical devices held by the medical device holder **20** to equipment that is not within the sterile field, such as for example to equipment that may be part of or positioned at the base **10** when the base **10** is not part of the sterile field.

Medical Device Holder

[0066] Medical device holder **20** is configured to hold a variety of medical instruments and with their transmission lines attached during, for example, a medical procedure. Various exemplary embodiments of a medical device holder according to the present disclosure enable secure holding of medical devices, for example in the sterile field, when not in use, such as prior to the surgical procedure when setup is occurring or during the surgical procedure when medical equipment is not actively in use. Thus, medical device holder **20** may prevent the medical instruments from rolling off or falling off medical device holder **20**. A length of medical device holder **20** may be sufficient to receive a shaft and back end housing mechanisms of various sized surgical instruments, including endoscopes, configured for use in telesurgical systems. A width of medical device holder **20** may be sufficient to hold two or more such instruments. In one embodiment, medical device holder **20** may have dimensions of 15 inches (38.1 cm)×30 inches (76.2 cm) for the overall lateral dimensions surrounding the face intended to hold the instruments. In other embodiments, medical device holder **20** may have a width ranging from 15 inches (38.1 cm) to 25 inches (63.5 cm) (inclusive); a length ranging from 20 inches (50.8 cm) to 30 inches (76.2 cm) (inclusive), and a thickness of up to three inches (7.6 cm) (inclusive).

[0067] With reference to FIG. 2A-1, an embodiment of a medical device holder **20** includes a tray **50** and a support frame **60**. Tray **50** includes a generally flat surface **52** with an upwardly protruding lip **55** around its outer perimeter. Thus, tray **50** may provide a flat working surface, while protruding lip **55** provides a barrier that prevents and/or minimizes the risk of instruments disposed on tray **50** from rolling off or falling off. In some embodiments, protruding lip **55** may be wide enough to include one or more recesses configured to hold a variety of objects such as, for example, clips, specimen containers, or transmission lines (not shown in FIG. 2A-1). In other exemplary embodiments, the tray **50** may have various surface features, such as recesses etc. that provide areas to hold items, such as sutures, scissors, etc. and/or portions of medical devices themselves. Various other embodiments of medical device holders with similar type tray structures having receptacles and other surface features are described further below. In some embodiments, the lip **55** may extend upwardly from the flat surface **52** up to three inches. It may be desirable to choose a height of the lip that accomplishes the desired outcome of preventing medical devices from rolling or falling off the tray while not posing an impediment from personnel removing the devices from the tray **50**.

[0068] Support frame **60** may be pivotably attached to tray **50**, e.g., at pivot couplings **61**, **62** shown in FIG. 2A-3, such that support frame **60** rotates relative to tray **50**, allowing support frame **60** to fold over tray **50** (FIG. 2A) or extend outwardly from tray **50** (FIGS. 2A-1, 2A-3). The tray **50** further includes one or more stops **58** (only one is illustrated in FIG. 2A-2) which permit the

support frame **60** to rotate relative to the tray **50**, but limit the range of rotational motion of the frame **60** relative to the tray **50**. In some embodiments, the stops **58** may be oriented so that the frame **60** cannot rotate past a horizontal position (e.g., past 180 degrees from the folded position), as illustrated in FIGS. 2A-2, 2A-3, that is, the support frame **60** and the tray **50** are in the same or parallel planes. In some embodiments, as illustrated best in FIG. 2A-3, support frame **60** may be of the same height as protruding lip **55** so as to provide a uniform height of the medical device holder whether in the folded or open, extended configuration. In the folded configuration shown in FIG. 2A-2, support frame **60** and protruding lip **55** may be in a nesting relationship such that support frame **60** surrounds protruding lip **55**. FIG. 2A-2 shows support frame **60** as being disposed on the outside of protruding lip **55** in the folded configuration. It is also contemplated, however, that support frame **60** may be disposed and nested on the inside of protruding lip **55** in the folded configuration. Moreover, the support frame **60** need not be nested with the protruding lip **55**, and the present disclosure contemplates embodiments in which the support frame **60** has a larger perimeter or smaller perimeter than the protruding lip **55**. Each of the support frame **60** and the tray **50** can have dimensions as discussed above with respect to medical device holder **20** of FIG. 1. [0069] In the folded configuration, medical device holder **20** may be configured for easy storage, for example, but not limited to, within a base **10** as discussed further below.

[0070] Support frame **60** may pivot outwardly relative to tray **50** in order to move from the folded configuration to the open configuration. In the open configuration, as shown in FIG. 2A-3, support frame **60** may be planar with regard to protruding lip **55** such that a top surface of support frame **60** is generally flush with a top surface of protruding lip **55**. Support frame **60** may form an opening **59** in the open configuration. As discussed further below, in the open configuration, the support frame **60** may support a sterile drape (such as the sterile drape **40** or other drape disclosed herein) or other flexible material so as to provide a flexible support pouch supported by the frame **60**. In an embodiment, by placing a drape over the entire medical device holder **20**, including both the tray **50** and the support frame **60** in the open configuration, the drape can provide a second holding region (within opening **59**) for medical devices and their associated transmission lines. Such embodiments are described further below with reference to FIGS. 11A, 11B, as well as FIGS. 2C and 2D.

[0071] Tray **50** and support frame **60** may be made of a variety of materials such as, for example, plastic, metal, metal alloys, such as stainless steel, and/or other medical grade materials. In addition, the tray **50** and support frame **60** may be made of materials that are sterilizable, such that for example, a sterile drape may not be needed to be placed at least over the tray **50** during the surgical procedure. Tray **50** may include the same or different material from support frame **60**. In some embodiments, tray **50** may be formed of a thin sheet of stainless steel over a thermoplastic body, and support frame **60** may be formed of a thermoplastic material.

[0072] FIGS. 2A-4 and 2A-5 show medical device holder **20** attached to an exemplary embodiment of arm **30** which, as discussed with reference to FIG. 1, can be used to support the medical device holder **20** and connect the medical device holder **20** to a base **10**. An arm **30** supporting the medical device holder **20** may allow for movement of the medical device holder **20**, and any medical devices held by medical device holder **20**, into a variety of positions during a surgical procedure, as discussed further below. Tray **50** and support frame **60** can help to minimize the risk of medical devices falling off medical device holder **20** while medical device holder **20** is being moved into the variety of positions, thus protecting and enhancing the life and sterility of the medical devices. Although the arm **30** in FIGS. 2A-4 and 2A-5 is shown as having various joints, such a configuration is exemplary and non-limiting, and an arm without such joints, as well as an arm in accordance with any of the embodiments described herein, can be used to support the medical device holder **20**.

[0073] In some embodiments, as shown in FIG. 2B, medical device holder **20** may include a first tray **53** and a second tray **54**, each with a protruding lip **56** extending from a flat, planar surface,

similar to the structure of tray 50. First tray 53 and second tray 54 may be pivotally connected such that first tray 53 and/or second tray 54 rotates relative to each other into a folded configuration or an open configuration in a manner similar to how the support frame 60 and tray 50 fold and open in the embodiment of FIGS. 2A-1 through 2A-5. For simplicity, only the open configuration is depicted in FIG. 2B. Protruding lips 56 may be of similar size and shape on first tray 53 and second tray 54. Similarly, the overall dimensions of the trays 53 and 54 can be similar to place the trays in a nesting relationship when in the folded configuration. Alternatively, though not shown, the trays 53 and 54 could have different dimensions. Additionally, a portion of one of the protruding lips 56 may separate first tray 53 from second tray 54 such that the portion of the protruding lips 56 provides a barrier between the trays 53, 54 in the open configuration.

[0074] In other embodiments, protruding lips 56 on first tray 53 and on second tray 54 may be a continuous and unitary member, and the first tray 53 and second tray 54 may not be pivotably coupled and rotatable to the folded configuration; instead the trays 53 and 54 may be a monolithic, fixed structure.

[0075] As shown in FIGS. 2C-1 and 2C-2, instead of having a first tray 53 and second tray 54 with the corresponding surfaces, a medical device holder 20 can have a similar configuration as described with respect to FIG. 2B, but with a first support frame 63 and a second support frame 64, each forming openings 69a and 69b and having structures similar to support frame 60 described in the embodiment of FIGS. 2A-1 through 2A-5. To provide surfaces for holding medical devices, a sterile drape or other sterilizable pouch, which can be permanently attached to the frames, can be provided to form wells 65, 66 (or other sized and shaped receptacles) extending from the support frames 63, 64 through the openings 69a and 69b. In some embodiments, the wells 65, 66 may be 8 inches×30 inches×7 inches deep. The drape or other sterilizable pouch disposed in the first support frame 63 may be the same or different from that disposed in the second support frame 64. One or more medical devices and/or cables may be stored within the wells 65, 66 thus formed.

[0076] Similar to the configuration described with reference to FIG. 2B, the overall dimensions of the support frames 63, 64 can be similar to place the frames in a nesting relationship when in the folded configuration. Alternatively, though not shown, the support frames 63, 64 could have different dimensions. Additionally, protruding lip 55 may separate support frames 63, 64 such that protruding lip 55 provides a barrier between the frames in the open configuration.

[0077] In one embodiment, an optional removable surface 67 may be disposed on a top portion of one of the wells 65 or 66, thus covering one of the openings 69a or 69b, as shown in FIG. 2D. Removable surface 67 may be configured to hold and secure one or more medical devices and/or miscellaneous items, similar to a tray. The removable surface 67 can be sized to provide a friction fit with the interior of the support frame 60, 63, 64, e.g., the interior surface of protruding lip 55, sufficient to hold medical devices without deforming the removable surface 67 or causing it to fall through the opening of the support frame. A top surface 68 of removable surface 67 may be planar or recessed slightly below a top surface of the support frame in which it sits. In the latter case, the support frame 64 provides a protruding lip surrounding the support surface 67, similar to the trays in the other embodiments described above. The removable surface 67 may be made of, for example, plastic, metal, paperboard, foam, rubber, etc. In addition, it may be provided with a textured surface (not shown) to provide better friction and prevent slipping and/or rolling of medical devices placed thereon. In some embodiments, the top surface 68 may include one or more widthwise and/or lengthwise depressions or grooves (not illustrated) to aid in organizing instruments, and inhibiting rolling of instruments on the top surface.

[0078] Drapes, pouches, and removable surfaces as described herein may be made of medical grade materials that are disposable or reusable. In the latter case, the materials should be sterilizable.

[0079] In some embodiments, as shown in FIG. 2E, a medical device holder 20 can include a first larger outer tray 73 (also referred to herein as “first tray” or “outer tray”) and a second smaller inner tray 74 (also referred to herein as “second tray” or “inner tray”) supported within the outer

tray **73**. In other words, the inner tray **74** can nest within the outer tray **73**. Each of the trays **73**, **74** can have a protruding lip **75**, **76**, respectively, extending from a flat, planar surface such as those described elsewhere herein. By way of non-limiting examples, the first and second trays can each be formed of metal or thermoplastic, with the inner tray **74** having dimensions generally the same as other trays described in the exemplary embodiments above and the outer tray **73** having sufficiently larger dimensions to hold the inner tray **74**. In an exemplary embodiment, the inner tray **74** has dimensions of the largest dimensions provided above with respect to other tray embodiments and the outer tray **73** has dimensions of 13 inches×20 inches. The arrangement of the trays allows the inner tray **74** to be removably held by the outer tray **73**. Protruding lip **75** may have a height substantially equal to or larger than protruding lip **76**. In an exemplary embodiment, protruding lip **75** can be slightly larger (e.g., 0.25 inches to 1 inches larger) than protruding lip **76** to accommodate the inner tray **74**. As discussed above, both first tray **73** and second tray **74** may be configured to secure and hold one or more medical devices, such as, for example, an endoscope and/or various surgical instruments. First tray **73** may also include one or more recesses **70** on either or both end of the tray to hold one or more tips and/or shafts of medical devices in a secure manner, such as via a friction fit, and/or to capture and route transmission lines connected to the cables. Thus, for example, the handle of a medical device can be disposed on second tray **74** and the tip of the medical device can be secured within recess **70** of first tray **73**. As shown in FIG. 2E, recesses **70** may provide additional stability to minimize and/or prevent the endoscope from rolling off of first and second trays **73**, **74**. Further, such recesses can protect delicate tips of medical devices when not in use. Recess(es) **70** can be formed integrally in or with outer tray **73** or can be part of a separate block **78** attached to the bottom or edge (or both) of the tray. In some embodiments, recesses **70** may have a V-shaped cross section, while in other embodiments may have a flared cross-section, such as the cross-section of a trumpet bell.

[0080] In the embodiment of FIG. 2E, inner tray **74** may be disposed on outer tray **73** without any separate attachment means. Alternatively, inner tray **74** may be secured to outer tray **73** with one or more attachment mechanisms such as, for example, an adhesive, a hook-and-loop-pile type fastener (e.g., Velcro), etc. Such attachment mechanisms may allow inner tray **74** to be removably attached to outer tray **73** without damaging or destroying the trays or the attachment mechanism. It is also contemplated that outer tray **73** includes a friction-inducing material in order to minimize any sliding of inner tray **74** relative to outer tray **73**.

[0081] In the embodiment of FIG. 2F, medical device holder **20** includes a center tray **80**, with support frames **83**, **84** having openings **85**, **86** on opposite sides of the tray **80**. This embodiment can function similarly to any of the embodiments described herein, as would be apparent to one having ordinary skill in the art. In an embodiment, the frames **83**, **84**, and tray **80** can be pivotably connected to provide a tri-fold configuration, including pivots and stops similar to those described with reference to FIGS. 2A-2 and 2A-3, or can be fixedly connected and integral (e.g., monolithic structure). As discussed above, one or more drapes or sterilizable pouches can be disposed in the frames **83**, **84** (see FIGS. 2C-1 and 2D) to provide wells to hold, e.g., medical devices and associated transmission lines. Additionally, medical instruments and other components may be disposed on tray **80**. In an embodiment, the entire medical device holder, including the tray **80**, can be draped.

[0082] In some embodiments, as shown in FIG. 2G, medical device holder **20** includes a tray **90** with a flat, planar surface and one or more depressions to form receptacles **91** therein. Receptacles **91** can be sized and shaped to receive one or more medical instruments. Thus, receptacles **91** may be a cavity on tray **90** in order to securely maintain the medical devices on tray **90**, and thus minimize the risk of the medical devices falling off tray **90**. As shown in FIG. 2G, medical device holder **20** may include additional storage compartments, such as recesses **93** sized and shaped to receive and route transmission lines of medical devices. Recesses **93** may be formed in a separate member from a protruding lip **95** of the tray or may be formed on a top surface of protruding lip **95**.

Additionally or alternatively, medical device holder **20** can include compartments **97** to store a variety of miscellaneous objects. Compartments **97** may be a unitary member with protruding lip **95**, or compartments **97** may be formed separately from protruding lip **95**. As shown in FIG. **2G**, protruding lip **95** may form an outer side surface of compartments **97**. Areas of the tray **90** also can be made of different materials. For example, foam, rubber or another deformable material may be used for the regions including the recesses **93** and compartments **97**, while the region having the receptacles **91** may be made of another material, such as stainless steel or other medical grade metal or plastic. In an exemplary embodiment, the main body region with the receptacles **91** can be sterilizable and reusable, and the end regions forming the compartments **97** and/or recesses **93** can be disposable and configured to be attached via friction fit or other removable coupling mechanisms on the main body portion.

[0083] FIG. **2H** shows a plan view of the medical device holder **20** of FIG. **2G** showing various surgical instruments **99** in the receptacles **91** and on the flat surface of the tray **90**, as well as other devices, which may be used during a surgical procedure, placed in the receptacles **91** and compartments **97**. Also shown in FIG. **2H** is how transmission lines connected to the instruments **99** can be received and routed through the recesses **93**, with tips of the instruments being placed to reside over the compartments **97**.

[0084] Receptacles in trays of medical device holders disclosed herein may be of varying sizes and shapes and may be configured to receive one or multiple medical devices and at least a portion of their cables. For example, as shown in FIG. **2I**, a tray **100** can have a receptacle or set of receptacles **101** configured to store the shafts of two or more endoscopes or other instruments, with the tips of the shafts secured within recesses **103** at one end of the tray sized to provide a friction fit to hold the tips of the shafts. Additionally, the handles of the endoscopes or other instruments may be disposed in a second receptacle **104**, or set of receptacles. As shown in FIG. **2I**, the receptacles **101** can be dimensioned to receive at least a portion of the transmission lines of the two endoscopes to facilitate transmission line organization.

[0085] Medical device holders in accordance with various embodiments can include a variety of other components that assist with organization, access to, and maintaining sterility of various devices that may be used during a surgical procedure. With reference to FIG. **2J**, a smaller, accessory tray **109** can be included that is configured to be removably and/or movably coupled to tray **110**. Such an accessory tray **109** can be useful to hold various smaller devices, such as for example, needles and scissors, that are useful in surgical procedures. The accessory tray **109** may be removably attached to be clipped or snap-fit into position over a portion of the tray, as shown in the upper illustration of FIG. **2J**. In another embodiment, the accessory tray **109** can be raised above the tray **110**, as shown in the lower illustration in FIG. **2J**. For example, one or more legs **112** can attach the accessory tray **109** to tray **110**. Optionally, the legs can be pivotable relative to the tray **110**, and optionally allow pivoting motion of the accessory tray **109** relative to tray **110**. By way of a non-limiting example, the legs **112** may form one or more four-bar linkages with the tray **110** and the accessory tray **109**. Therefore, the legs **112** may allow accessory tray **109** to pivot from a first configuration disposed over and adjacent tray **110**, to a second configuration in which the accessory tray **90** is vertically displaced relative to tray **110**. Legs **112**, accessory tray **109**, or both, may optionally include locks or the like to secure the tray **109** in the lower orientation, the raised orientation, or both. One of ordinary skill in the art would appreciate a variety of manners in which the accessory tray can be attached to and located relative to any of the trays or support frames of medical device holders described above, without departing from the scope of the present disclosure. Accessory tray **109** may include one or more of the additional features described with reference to the exemplary embodiment of FIG. **2G-2I**, such as various recesses, compartments, pouches, receptacles, and may be made of, for example, plastic, metal, paperboard, foam, rubber, etc., including sterilizable materials if intended to be reusable.

[0086] In other embodiments, various other storage devices can be included as part of a medical

device holder **20**. As shown in FIGS. **3A-1** and **3A-2**, a holster **300** having a receptacle sized and shaped to receive a shaft of a surgical instrument, including an endoscope, can be attached to the outer periphery of the medical device holder **20**. In one embodiment, holster **300** is clipped onto the protruding lip of a tray or a frame, with the receptacle oriented vertically. Holster **300** can be removably or fixedly secured to medical device holder **20**. As shown in FIG. **3A-1**, holster **300** may be used to secure medical instruments in a vertical position. As shown in FIG. **3A-2**, holster **300** may include an elongated sheath or cover **302** extending downward from the holster, of a length sufficient to cover and protect the surgical instrument, including a distal tip of the surgical instrument.

[0087] Other storage compartments include removable or fixedly coupled cartridges **310**, as shown in FIG. **3B**. Such cartridges **310** may be, for example, clipped onto a protruding lip of a tray or the frame. A variety of objects may be stored within cartridges **310**, such as, for example, sutures, gauze, etc. Cartridges **310** also can be used to store liquids, such as liquids to clean various medical devices, such as endoscopes, for example, during a medical procedure.

[0088] A medical device holder in accordance with various embodiments also may include one or more pouches (e.g., pockets) **320** that hang from an outer periphery of the frame or tray of the medical device holder. As shown in FIGS. **3C-1** and **3C-2**, pouches **320** can be removably attached or fixedly secured to a side or other edge surface of a tray **350**, for example, to the protruding lip, or support frame (not shown). In some embodiments, pouches **320** are attached to a tray or support frame using an adhesive, a clip, a snap-fit mechanism, or other suitable attachment device. As shown in FIGS. **3C-1** and **3C-2**, pouches **320** hang downward from the tray **350**. Pouches **320** can be sized and shaped to provide convenient storage for a variety of objects including, for example, transmission lines of medical devices, as depicted in FIG. **3C-1**. Pouches **320** may each be connected together or may each form separate members when attached to medical device holder **20**. In the example of FIGS. **3C-1** and **3C-2**, three pouches are shown, but any number is within the scope of this disclosure. The pouches **320** may all be the same size, or the pouches **320** may be differently sized. Pouches **320** can be made of flexible and durable material such as, for example, polyethylene (PE), polypropylene (PP), polystyrene (PS), or polyvinyl chloride (PVC). In one embodiment, the tray and/or support frame of the medical device holder and the pouch can be sterile or formed of a material which is sterilizable. In another embodiment, the pouch can be packaged as a sterile, disposable component and configured to be attached to a reusable and sterilizable tray and/or support frame. In some embodiments, a pouch may be up to **11** inches deep, such as, for example, 6-8 inches deep, and have a width up to the length or width of the tray from which it is hung.

[0089] Medical device holder **20** also can include or be configured to receive a variety of transmission line routing features. For example, various clips, hooks, and the like can be used to organize and route transmission lines of medical devices. FIG. **4** provides an example of a hook **330** that may be used to route transmission lines, which can be placed in pouches, as shown, or not. Hook **330** may be used to prevent sagging, tangling, and general disorganization of transmission lines when medical instruments are held by a medical device holder. Various other transmission line routing structures are discussed further below.

[0090] It is within the scope of this disclosure that various aspects of the above-discussed embodiments of medical device holder **20** may be combined. Thus, the features of each of these embodiments may be combined with features of the other embodiments. Various features of the different embodiments are not mutually exclusive and, instead, can be combinable as those have ordinary skill in the art would understand

[0091] Medical device holder **20** provides enhanced instrument and cable organization compared to conventional surgical instrument trays, such as Mayo stands. For example, the flexible support wells, trays with lips and various receptacles, tip and transmission line recesses, pouches and other storage compartments and devices provide for easy storage of the medical devices, associated

transmission lines, and other devices that may be used during a surgical procedure. Moreover, the medical device holders and related systems can provide for an overall efficient workflow prior to and during a surgical procedure. Additionally, medical device holders and related systems can allow medical devices to be easily switched in and out of use during a surgical procedure.

[0092] Further, medical device holders in accordance with various embodiments are configured to minimize the risk of medical devices falling off the medical device holder, thus reducing the risk of potential damage to such medical devices and increasing the lifespan of thereof.

Support Arm for Medical Device Holder

[0093] As mentioned above, medical device holder **20** can be fixed to or removably coupled to an arm **30** extending from a base **10**. In embodiments wherein the medical device holder **20** can be removably coupled to the arm **30**, the medical device holder **20** can be provided sterilized, and additionally may be sterilizable (e.g., autoclavable) to provide for reuse. Providing a removable medical device holder **20** that is sterile or sterilizable may allow for simplicity in the setup of the operating theatre, and potentially reuse of the medical device holder **20** over multiple procedures. Further, a sterile drape over the medical device holder **20** may not be required in such an embodiment. It also may facilitate post-procedure reprocessing of medical devices held on the medical device holder **20** by allowing the medical device holder **20** with medical devices held thereon to be removed from the arm **30** and moved to a remote location for cleaning. Moreover, removal of the medical device holder **20** from the arm **30** may facilitate storage of the arm and any base to which it is attached.

[0094] In addition, permitting the medical device holder **20** to be removable from arm **30** can enhance the use of the medical device holder **20**. For example, a user may disconnect medical device holder **20** from arm **30** in order to move medical device holder **20** to another position in the operating room that is remote from arm **30** and base **10**.

[0095] All of these various features can result in creating an overall efficient workflow.

[0096] FIGS. 5A and 5B show an embodiment in which an arm **530** attaches to a base **510** at one end and includes a sterile adapter member **540** at a distal end of the arm **530** opposite the end that attaches to the base **510**. Arm **530** may be used as arm **30**, and base **510** may be used as base **10**. Arm **530** and base **510** can have a variety of configurations with the embodiment depicted in FIGS. 5A and 5B being nonlimiting; additional details and embodiments of arms and bases, and connections thereof, in accordance with the present disclosure are discussed below. By including a sterile adapter member **540** on the arm **530**, a sterile medical device holder **520** (e.g., from a package or otherwise sterilized prior to use) can be coupled to the sterile adapter member **540** by way of integrated attachment features (e.g., clips, grooves, undercuts, etc.) or by sterile bolts, screws, and/or other fasteners provided and used to secure the medical device holder **520** to the arm **530** (see FIG. 5B). Medical device holder **520** is shown as a tray in FIG. 5B but any of the medical device holder configurations described herein can be used. Thus, in the embodiment of FIGS. 5A and 5B, sterile draping of only the arm **530**, and the sterile adapter **540**, may be needed.

Alternatively, the sterile adapter **540** with the fasteners can be provided as a separate sterile component along with the medical device holder **520**, thus requiring draping only of the arm **530**. In either case, the fastening mechanism(s) used to attach the medical device holder, sterile adapter, and arm can be such as to operate through the drape.

[0097] Other connections of the medical device holder **20** to an arm **30** are envisioned as well. For example, whether for the embodiment of FIGS. 5A and 5B or other embodiments, including when the medical device holder **20** is fixedly coupled to arm **30** (in which case draping of the arm **30** and medical device holder **20** is envisioned), the medical device holder **20** can be rotatably or pivotably coupled to the arm **30** to allow some range of movement relative to the arm **30**. In this way, the medical device holder may be moved into a variety of orientations and positions to facilitate access to medical devices on the medical device holder **20** or clearing of areas within the sterile field by moving the medical device holder **20** out of the way. In an exemplary embodiment, a medical

device holder can rotate in a plane of the medical device holder and relative to a distal end of the arm over a range from 0 degrees to 360, and any sub-range therein. In an exemplary embodiment, the coupling mechanism between the arm and the medical device holder may allow for rotation in either direction. Moreover, the rotatable coupling mechanism can provide sufficient friction so that the angular position of medical device holder **20** relative to the arm **30** is maintained as the medical device holder **20** assumes a variety of positions. By way of a non-limiting example, a coupling mechanism may include a rotating shaft (not illustrated) extending from the arm, supported by rotary bearings (not illustrated) in the arm, and the sterile adapter member may include an undercut or flexible tabs (not illustrated) which permits the sterile adapter member to snap onto the rotating shaft. Those having ordinary skill in the art would recognize various other rotatable coupling mechanism could be used to couple the medical device holder to the arm.

[0098] Additionally, medical device holder **20** may be coupled to arm **30** so as to provide a slight tilt of the medical device holder **20** (e.g., along a pitch and/or roll axis of the medical device holder), relative to the arm **30**. Thus, medical device holder **20** may move upward and downward relative to arm **30**. Providing the ability to slightly tilt may allow the medical device holder to absorb unintended bumps or movements that may otherwise cause the medical device holder to become dislodged from the arm or affect the securement mechanisms.

[0099] While the present disclosure contemplates the use of arms with a single link member and relatively simple structure, such as generally a beam extending from a base **10**, various exemplary embodiments contemplate an arm that is jointed and can provide a range of motion and positioning, relative to both the base and the medical device holder, that enhances user access, flexibility of use, and reach into the sterile field, while maintaining secure holding of medical devices and connection of medical devices to auxiliary equipment outside the sterile field, as well as organized cable routing. Moreover, various arm configurations can provide safety features that may minimize the risk of damage and/or undesirable or unexpected movements of equipment within the operating theatre.

[0100] In an exemplary embodiment, as shown with reference again to FIGS. 5A and 5B, and to FIGS. 6A-6C, arm **530** can be configured to have an extended and a retracted configuration relative to the base **510**. In some embodiments, the base **510** comprises a compartment **512** and the arm **530** is received within the compartment **512** in a retracted configuration, as shown in FIG. 6C and described further below. In some embodiments, the base **510** may comprise other portions in addition to the compartment **512**, as illustrated in FIGS. 6D and 6E.

[0101] As shown, arm **530** includes a first link member **560**, a second link member **563**, and a third link member **565**. The first and second link members **560**, **563** and the second and third link members **563**, **565** can be rotatably coupled at first and second joints **570**, **573**, respectively. Further, as described above, a medical device holder can be connected by a rotatable coupling at a distal end of the third link member **565**, with the sterile interface adapter connection shown and described in FIGS. 5A and 5B being one nonlimiting embodiment of such a connection; other connections between third link member **565** and a medical device holder can be used.

[0102] In some embodiments, the arm **530** is configured to allow repositioning of the holder without risk of the transmission lines becoming detached from the base **510** or the medical devices being pulled off of the holder due to a change in cable distance between the transmission line ends, regardless of the position of the medical device holder. It may therefore be advantageous for the first link member **560** to have a fixed position after deployment (discussed in greater detail below) into the extended position, with the transmission lines being routed through a retaining device, e.g., a clip, at the end of the first link member **560** (also discussed in greater detail below). The remaining link members can be rotationally coupled to each other with transmission lines routed through guide features attached to one or more of the rotational joints so as to minimize the change in length of the transmission lines during repositioning of the arm and medical device holder.

[0103] The range of motion of the arm **530** may be limited to assist in minimizing inadvertent

contamination of the sterile medical device holder and sterile portion of the arm 530 and transmission lines, for example, by coming into contact with the non-sterile base. The fixed position of a first transmission cable guide device, e.g., clip, after the first link member is deployed, can be selected to enforce adequate distance between the non-sterile base and the sterile portion of the transmission lines, so that a draped arm can be manipulated by a sterile user without fear of contamination. Additional features, described elsewhere herein, may also be used to enforce this, such as, by way of non-limiting examples, locking out the first link member 560 when deployed to prevent the first link member 560 from sliding back (retracting back) into the base 510, limiting the rotational range of motion of the second link member 563 when draped, and including in the system mechanisms which tend to promote moving the holder is fully extended before providing the ability to drape the holder and arm 530.

[0104] As discussed further below, first link member 560 connects to base 510 such that first link member 560 can move in translation relative to base 510. This can allow the arm 530 to extend and retract in translation toward and away from the base 510, as illustrated in the positions of the first link member 560 shown in FIGS. 6A-6C. Additionally, in various exemplary embodiments, a pivoting connection may be used such that at least when first link member 560 reaches an extended configuration from the base 510, as shown in FIGS. 6A and 6B, it can also pivot to some degree of rotation relative to the base 510 (e.g., about an axis parallel to the axes of rotation of the first and second joints 570, 573).

[0105] Thus, first link member 560, second link member 563, and third link member 565 are configured to be folded relative to each other to provide a compact configuration for the arm 530 when not in use, for example, to be retracted into base 510, as discussed further below, and also to provide a range of orientations of the arm 530 and the medical device holder as those having ordinary skill in the art would appreciate. FIG. 6A shows first link member 560, second link member 563, and third link member 565 in an extended position (such that arm 530 extends outward from base 510). In this extended position, second link member 563 is pivotable relative to first link member 560 around first joint 570, while the first link member 560 is fixed relative to the base 510. Second link member 563 is pivotable relative to third link member 565 around second joint 573. Due to the pivoting movement of first, second, and third link members 560, 563, 565 about the first and second joints 570, 573, the arm 530 may assume a variety of positions. Reference is made to FIG. 5B showing an example of an orientation of the link members at different rotational positions about the first and second joints 570, 573. The first and second joints 570, 573 may be modest friction joints to hold positions once moved, but easy enough to pivot by a user wanting to move the arm to differing configurations. In an exemplary embodiment the range of motion about the joint axes can be about ± 180 degrees or more. In an embodiment, one or more of the joints can comprise a shaft attached to an end of a link, e.g., second link 563, extending vertically; one or both of the adjoining link(s), e.g., the first link 560 and the third link 565, include rolling bearings secured between a vertical bore formed in the link and a cap, with a hole in the cap for one end of the shaft to protrude.

[0106] Each of the first and second joints 570, 573 can alternatively have locking features (not shown) to separately lock into place in order to separately lock the position of first, second, and third link members 560, 563, 565, relative to the adjoining link(s) and/or base 510. Thus, for example, first joint 570 may be locked in order to lock first link member 560 in a first position while second link member 563 may pivot relative to first link member 560.

[0107] In some embodiments of a locking feature, the first link member 560 may include a horizontally pivotable lever that extends into and engages a detent formed in the base 510 when the first link member 560 is fully extended from the base 510 and the second link member 563 is rotated away from the folded, stow position. Rotating the second link member 563 from the folded position relative to the first link member 560 releases the lever and permits it to engage into the detent in the base 510, thus locking the first link member 560 in place relative to the base 510.

Depressing the exposed end of the lever, either manually or by returning the second link member **563** to the folded position relative to the first link member, pivots the other end of the lever out of the detent, thus unlocking the first link member **560** from the base **510**.

[0108] In some embodiments, a hard stop feature is included, that is only activated when the arm link members are deployed and the sterile drape is attached. Such a hard stop feature may prevent the second link member **563** from rotating back toward the stowed (fully folded) position more than 90 degrees relative to the first link member **560** (see FIG. **6E**). A hard stop feature may minimize accidental breach of sterility if the medical device holder is moved toward the side of the base **510** or if the second link member **563** is moved to the stowed position. A hard stop feature also may prevent deactivation of the first link member detent, described above, which keeps the first link member **560** in the extended position during use.

[0109] When first, second, and third link members **560**, **563**, **565** are fully extended, arm **530** can have an overall length ranging from about 20 inches (50.8 cm) to about 40 inches (101.6 cm), for example about 30 inches (76.2 cm) to about 35 inches (88.9 cm). Such an extension length may allow arm **530** to extend a sufficient distance from the base **510**, which may be located in a nonsterile field, to position an attached medical device holder safely within the sterile, surgical field. Thus, such an extension length may allow arm **530** to freely move within the sterile field, providing workflow flexibility to the user, while maintaining an adequate distance from the nonsterile environment, thus protecting the sterile environment from the nonsterile environment.

[0110] FIG. **6B** and **6C** illustrate an exemplary process of moving arm **530** from the retracted position (FIG. **6C**) to a partially extended position (FIG. **6B**) to allow further positioning of the arm **530** relative to the base **510**, such as in a fully extending position shown in FIG. **6A**. In order to maneuver arm **530**, first link member **560** can be moveably coupled to, for example, a slide rail **210** connected to the base **510**. The moveable coupling allows the first link member **560** to translate, e.g., slide, relative to the base **510** and to be moved away from the base **510** as shown in FIG. **6B**. By way of a non-limiting example, a bearing block **567** may be mounted to the first link member **560**, e.g., to a top surface of the link member, which contains ball bearings (not illustrated). The bearings roll against the rail **210**, portions of which extend through the bearing block **567**, and other portions of the rail **210** are mounted to the base **510** to provide low friction linear motion with high stiffness in the other directions. However, other couplings are envisioned that can provide translational movement of the first link member **560** relative to the base **510**, with the sliding rail being a non-limiting embodiment. In other embodiments, the entire arm **530** could be pivoted and pitched down from a vertical orientation to a final extended horizontal position, or yawed horizontally 90 degrees to the final position, or may include a four-bar linkage in the place of the linear slide rail described above.

[0111] In the embodiment shown, the base **510** includes a compartment **512** in which the arm **530**, when in the folded configuration, can be housed when the arm **530** is in the retracted position (FIG. **6C**). To slide the arm **530**, the distal end of the third link member **565** can be pulled, causing the first link member **560** to slide along the slide rail **210** in the direction **514** from the configuration in FIG. **6C** to that in FIG. **6B**. The arm **530** generally can move to the position in FIG. **6B** while in the folded configuration. Thus, FIGS. **6B** and **6C** illustrate a folded configuration in which the arm **530** can move in translation relative to the base **510**. FIG. **6C** illustrates a fully retracted configuration such that first joint **570** is disposed within base **510**. FIG. **6B** illustrates a partially extended position such that first joint **570** and second joint **573** are in a position to allow unfolding of the first, second, and third link members **560**, **563**, **565** relative to each other, for example, in the embodiment shown the first and second joints **570**, **573** are disposed exterior of base **510**. In the partially extended position, a proximal end portion of first link member **560** may still be disposed within base **510**.

[0112] As shown in FIGS. **6A-6C**, arm **530** includes a handle **590** or other grasping mechanism that can be used to allow a user to pull the arm **530** in direction **514**. In some embodiments, handle **590**

may be disposed on third link member **565**. It is also contemplated that the movement of first link member **560** from the first retracted position of FIG. **6C** to the partially extended position of FIG. **6B** can occur in an automated manner, rather than manually, such as via motors driving the movement of the first link member **560** along the slide rail **210** or via another motor-driven movement mechanism (e.g., worm gear etc.). In that case, handle **590** may be provided to override a malfunction of such automated movement.

[0113] During the extension process of arm **530**, and once first link member **560** has assumed the partially extended position of FIG. **6B** and the second link member **563** has been partially rotated relative to the first link member **560** to engage the locking lever (discussed above), the first link member **560** may be locked into position to keep from translating along the slide rail **210** and back to the retracted position of FIG. **6C**. Such locking also can prevent first link member **560** from moving horizontally or vertically relative to base **510** until arm **530** is in the extended position.

[0114] To further extend the arm from the partially extended position of FIG. **6B** to the fully extended position of FIG. **6A**, the third link member **565** can continued to be pulled in the direction of extension, which begins to cause the second link **563** member to rotate about first joint **570** so that second link member **563** becomes unfolded from first link member **560**. Simultaneously or subsequent to the unfolding of second link member **563**, the third link member **565** can be rotated about second joint **573** causing third link member **565** to unfold relative to second link member **563** so that third link member **565** becomes unfolded from second link member **563**. Eventually, arm **530** can thus reach the fully extended configuration shown in FIG. **6A**.

[0115] It is also noted that second link member **563** and third link member **565** need not be unfolded in order for arm **530** to assume the partially extended configuration, in which configuration the arm **530** can provide a sufficient distance into the sterile field as has been described above. In other embodiments, the arm **530** cannot be draped in a partially extended position, requiring the user to unfold the three link members to drape the arm **530**. Once the arm **530** is draped, cable guide (discussed elsewhere herein) engage one or more hard stops that require a minimum angle, e.g., a 90-degree angle, between the first link member **560** and second link member **563** (as discussed elsewhere herein).

[0116] In order to move arm **530** from the extended position of FIG. **6A** to the retracted position of FIG. **6C**, the reverse order of operation as described above can occur. For example, third link member **565** can be pushed in a direction toward the base **510** and rotated to cause rotation of second link member **563** about first joint **570** in the opposite direction as the rotation in the unfolding step. Once fully folded, third link member **565** can continue to be moved toward the base **510**, causing the eventual movement of the entire arm **530**, in the folded configuration, along the rail **210** to the retracted position of FIG. **6C**. As mentioned above, it is envisioned that any of the movements of the arm **530** can be automated using motors and controls.

[0117] Referring now to FIG. **6D** and **6E**, the three-link jointed arm **530** holding a medical device holder **520**, which in FIGS. **6D** and **6E** has an embodiment similar to that of FIGS. **2A-1** through **2A-5** but is not limited to such a configuration, is shown to illustrate a range of motion relative to an exemplary base **510**. FIG. **6D** shows the arm **530** in the fully extended configuration, as described above with reference to FIG. **6A**. As depicted, with the attachment of the medical device holder **520** to the arm **530**, a considerable reach into the sterile field from the base **510** can be achieved. From such a position, FIG. **6E** further illustrates how the second link member **563** can be rotated to roughly 90 degrees relative to the first link member **560** about first joint **570**, and the third link member **565** can in turn be rotated about 90 degrees relative to the second link member **563** about second joint **573** to move the arm **530** and the medical device holder **520** generally away from the surgical field and tucked back, yet outside of the base **510**. Such an arrangement again can provide ease of use of the entire system before, during, and after a surgical procedure to provide a range of stable motion of the medical device holder in positions both within the sterile field and outside of it. The further incorporation of one or more hard stops (discussed above), which can be

activated when a drape is installed, enforces the rotational placement of the link members into a configuration that helps avoid contact between the sterile medical device holder **520** and the non-sterile cart or base **510**. Additional rotary range-of-motion hard stops may enforce arm positions toward the side of the cart opposite to that illustrated in FIG. **6E**, which may minimize the chance of breach of sterility in the extreme position of FIG. **6E**.

[0118] In the embodiment of FIGS. **6A-6E**, the arm **530** shows link members **560**, **563**, **565** as having central longitudinal axes that remain generally parallel to each other. Additionally, the central longitudinal axis of second link member **563** is disposed below the central longitudinal axis of first link member **560** and of third link member **565** in order to provide a greater degree of motion to arm **530** and to allow arm **530** to be stored in the retracted position within base **510**. In some embodiments, the central longitudinal axis of first link member **560** and of third link member **565** may be coplanar and remain at the same height regardless of the pivoting orientation of first, second, and third link members **560**, **563**, **565**.

[0119] In various embodiments, it may be desirable to allow additional movements of an arm, to further customize and add position and orientation flexibility of a medical device holder. FIG. **7** illustrates another embodiment of an arm having a three link jointed configuration. Specifically, FIG. **7** illustrates an arm **730**, which may be used as the arm **30**. The arm **730** has a first link member **760**, a second link member **763**, and a third link member **765**, similar to the arm **530** described with reference to FIGS. **6A-6E**. In addition, the arm **730** further includes a coupling of the third link member **765** (that can be a parallelogram linkage structure) that allows the third link member **765** to not only rotate around the joint **773** (i.e., around axis of rotation **774**), but also to move up and down over an angle α between a position in which the longitudinal axis of the third link member **765** is generally parallel to that of the second link member **763** and a position in which the longitudinal axis of the third link member **765** is angled relative to the longitudinal axis of the second link member **763**. More specifically, the third link member **765** may be moved from a first parallelogram orientation to a second parallelogram orientation which is different from the first parallelogram orientation. The first parallelogram orientation may be a rectangular orientation of the third link member **765**, and the second parallelogram orientation may be non-rectangular. The parallelogram linkage can permit this movement without altering an orientation of the distal end of the arm **730** so as to maintain the medical device holder (not shown) in a generally planar orientation (e.g., the surface generally parallel to the ground) as the distal end of the third link member **765** moves vertically. The range of motion of the vertical movement of the third link member **765** can range from ± 5 inches (12.7 cm) to ± 10 inches (25.4 cm), for example about ± 8 in (20.3 cm), measured as movement of the distal end of the third link member **765**. In other words, the third link member **765**, in addition to being rotatable about axis **774** via joint **773**, can rotate through angle α . By way of a non-limiting example, a four-bar linkage may be part of, or used as, third link member **765**. In such an embodiment, an upper link of third link member **765** is attached at both ends via pins **767** and can pivot around plain bearings to rotary joint housings, and a bottom link, parallel to the top link, of third link member **765** is also attached to and can pivot about the rotary joint housings via pins **769**. If the distance between pins **767**, **769** is the same in both housings, and the length of the top and bottom links are equal, the medical device holder (not shown) will remain parallel to the ground. In some embodiments, the range of angle α is ± 40 degrees, but other values for the range of this angle may also be used.

[0120] Arm **730** also can include a motor to move third link member **765** vertically to the position generally parallel to second link member **763** through the range of angular motion denoted by angle α . The motor may be disposed on arm **730**, or arm **730** may be remotely connected to the motor. For example, a DC brush motor may be operably connected to cause the angular motion of the third link member **765**. A vertical motion switch **200** may be accessed by a user in order to activate the motor to move, for example, third link member **765** from the horizontal position to the inclined position illustrated in FIG. **7**. Switch **200** may be operably coupled to the motor and can be

disposed on arm **730**. Alternatively (not shown), a switch may be provided on a portion of the medical device holder and configured to be operably coupled to a motor that raises the third link member **765**.

[0121] In yet other embodiments, it is envisioned that the entire arm **30** can have a limited range of vertical movement over an angle relative to the base. This can be useful, for example, if a patient bed were to raise up and hit the arm and/or if the medical device holder were to get caught under something or a force applied from above. Configuring the arm to allow some tolerance in movement vertically can enhance overall stability of the system in various potential use conditions.

[0122] Some embodiments may include a mechanism which permits a limited range of vertical movement of the arm over an angle relative to the base. By way of a non-limiting example, one such mechanism may include a yoke having a horizontally oriented bore. The yoke is attached to the bearing block, e.g., bearing block **567**, which cooperates with the linear slide rail **210**, discussed above with reference to FIGS. **6A-6C**. The proximal end of the first link member is pivotably connected to the yoke, instead of being rigidly connected to the bearing block, by a horizontal pin so that the first link member can rotate in a pitch direction with respect to the bearing block and, therefore, relative to the base, but is constrained in all other directions. The first link member is counterbalanced or otherwise forced upward by a compression spring positioned on the proximal side of the yoke's pivot and held against a hard stop formed on the base. This allows movement of the arm downward against the force of the spring when excessive downward loads are applied to the medical device holder or to the arm. In some embodiments, a sensor may be provided to detect this movement and alert the user of the excessive load, e.g., by closing a circuit to sound a buzzer, energize a light, or the like. This could prevent overbalance of the base due to an operating room table moving down and making an undetected collision with the arm. In yet other embodiments, a weight may be used to counterbalance the arm in the place of or in addition to the compression spring. In other embodiments, a second pivot may be provided with a counterbalance at the proximal end of the arm to perform this same function for excessive loads that are applied upward.

[0123] In some embodiments, arm **30** may include a counterbalancing mechanism to offset any weight placed on arm **30**. Thus, arm **30** may maintain its vertical position even when a weight is deposited on arm **30**, for example, if a user were to lean on the top of arm **30**. In some embodiments, arm **30** may be counterbalanced for about a 30 lb payload with the arm fully extended. In general terms, this payload must be large enough to not falsely trigger, but should be much lower than that needed to overbalance the cart or base. The load causing overbalance of the cart or base is dependent on the distance from the point of application of the load to the base, so the most extended orientation of the arm can be used to determine a payload above that which is normally anticipated and thus should be tolerated before movement of the arm relative to the base.

[0124] In some embodiments, arm **30** may include a back-drivable mechanism to inhibit, and may prevent, arm **30** from inadvertently moving horizontally. For example, arm **30** may be inhibited or prevented from moving horizontally if a user were to accidentally fall into arm **30**. Such may inhibit or prevent arm **30** from inadvertently moving and causing medical instruments to fall off medical device holder **20**. The back-drivable mechanism of arm **30** may be accomplished by the structures of the first and second joints **570**, **573** for the various three link jointed arm embodiments described herein.

[0125] The couplings at the joints between the link members of the arm in various embodiments described herein can have a variety of features that permit smooth and controlled operation. It is desirable to have joint structures that allow for the medical device holder, and arm, to be relatively easily moved without drifting. For example, as shown in FIG. **8**, the joint coupling can be a SCARA (Selective Compliance Assembly Robot Arm) joint that includes a friction or brake pad **780**, a rotor, which may be single sided, **783**, and a spring **784**, which may be a wave spring. The brake pad **780** generates controlled rotational friction when pushed against the rotor **783** by the wave spring **784**. Spring **784** allows the friction to be maintained at an approximately consistent

level independent of brake pad wear. The brake pad **780** is fixed to one link member via fasteners, and the rotor **783** is constrained against rotation relative to the interfacing link member but can move axially toward and away from the brake pad to compensate for brake pad wear. In a non-limiting example, pins are provided in the link member which extend through corresponding mating holes in the rotor.

[0126] FIG. **9A** depicts another exemplary embodiment of an arm having three link members coupled by joints in a manner similar to that described with respect to the embodiment of FIGS. **6A-6E**. Specifically, FIG. **9A** illustrates an arm **630**, which may be used as the arm **30**. In the embodiment of FIG. **9A**, the first link member **660** attaches at a side of the link member **660**, rather than its top, to a slide rail **610** to allow translational movement of the arm relative to the base (not shown). A breakaway mechanism (not shown) also can be provided between the first link member **660** and the bearing block mounting the first link member **660** to the slide rail **610**. Also, the third link member **665**, having the parallelogram linkage **695**, can include a back-drivable lead screw (not shown) to provide protection against overload of the arm by payload of the medical device holder and/or lifting of the arm from the underside, such as by a patient or operating table being lifted into the medical device holder or arm from underneath. In some embodiments, backdriving may be achieved by providing a lead screw and a mating lead nut, with the lead nut including ball bearings to create a low friction interface with the lead screw. Varying the pitch and motor friction to rotate freely upon exertion of a specific axial load allows backdriving.

[0127] Further, in the embodiment of FIG. **9A**, and with additional reference to FIG. **9B**, it is contemplated that the medical device holder can be coupled to the distal end of the arm **630** via a connecting member **693** that serves as a floating “hand” to permit vertical movement of the medical device holder relative to the second link member **663** while the medical device holder remains horizontal, that is, level with the floor. This may be useful in situations where the patient or operating table rises up to place a significant force under the medical device holder, or if the user inadvertently moves the device holder downward into the patient by activating the motor. The connecting member shown in FIG. **9B** includes a first member **685** connected to a second member **686** via needle bearings **690**. A parallelogram linkage **695** connects first member **685** to second member **686** in order to provide a range of motion of about 30 mm to about 45 mm between the first and second members **685**, **686**. Second member **686** may be mounted and attached to arm **630**, for example at the distal end joint of the arm **630** at laterally extending flange **697**, and first member **685** can be attached to medical device holder (not shown). This can allow the connecting member **693** and medical device holder to pivot around the joint at the end of the arm **630**.

Parallelogram linkage **695** provides a connection between first and second members **685**, **686** in order to connect arm **630** with medical device holder **520** (not shown). In some embodiments, the connecting member **693** can be a sterile adapter member that attaches a medical device holder to the arm, for example, in an embodiment in which the tray is a sterile tray as discussed above. In this manner, the connecting member **693** constrains the link members to only move vertically and remain level.

[0128] In an embodiment, a portion of the connecting member **693** can be a contact switch. In this way, if something, such as a patient on an operating table or part of the table is under the medical device holder and comes into contact with the connecting member **693**, the action of the parallelogram linkage **695** can be actuated to slightly lift the medical device holder upwardly relative to the arm in a gentle controlled motion to avoid damage to sensitive components of the arm or to the patient. In an exemplary embodiment, the connecting member **693** can be designed to move once a payload of the entire medical device holder and its devices supported thereon is on the patient. The parallelogram linkage **695** is passive and thus can be moved at any speed. In an embodiment, however, the vertical movement permitted by the parallelogram linkage **695** is from 30 mm-45 mm of travel in about 1-1.5 seconds.

[0129] In various exemplary embodiments, arms (such as any of the arms **30**, **530**, **630**, and **730**)

may be made relatively lightweight so that the maximum payload is a predetermined amount, about 90 N in an exemplary embodiment, including the weight of medical device holder **20**. Further, one or more joints may include a breakaway mechanism. The coupling mechanisms between an arm and a base also may allow for the arm to be provided as field-replaceable units.

[0130] An arm **30** in accordance with various exemplary embodiments disclosed herein may also include one or more sensors to monitor the position of arm **30**, as well as various automated control features to permit adjustments or movements of the arm and/or medical device holder so as to provide enhanced safety.

[0131] Arm **30** may include various electronics components. For example, FIGS. **10A-1** to **10A-3** illustrate an embodiment of an arm **830**, which may be used as the arm **30**. As shown in FIG. **10A-3**, the arm **830** comprises a first link member **860**, a second link member **863**, and a third link member **665** coupled together by joints, similar to the embodiment of FIG. **9A**. The arm **830** comprises various electronics components. Such electronic components may include a stow momentary switch **802**, which may be disposed underneath the drape when the drape is placed over the arm. The electronic components may also include contact switches to detect contact between various portions of the arm, medical device holder, and external objects. For example, contact switch **804** may be located on the underside of the wrist and parallelogram to detect contact between the underside of the medical device holder and distal portion of the arm. A contact switch **806** may be located to detect contact between the arm or medical device holder and an external object. Other components include an LED **808** for notification of contact with the underside of the medical device holder and/or distal portion of the arm with an external object, such as a patient, a DC brush motor **810** (low duty cycle, low usage), an electronic brake **812** at the motor (normally on), an absolute position sensor **814**, and additional switches for tray position and/or drape presence.

[0132] More specifically and with continued reference to FIGS. **10A-1** and **10A-2**, in some embodiments a stow momentary switch **802** includes a snap-action micro-switch that is triggered only when the medical device holder is fully extended, and is used to prevent motorized vertical movement that could cause collisions and damage between the third link member and the holder. Concerning the switches for detecting contact between the underside of the holder and the distal portion and the arm, a most distal switch **806** is positioned at the floating hand (described above) and is positioned to detect contact between the holder and an object. The next two contact switches **804** are positioned to detect contact between distal portions of the arm that could come into the sterile field. The LED **808** lights up to alert the user of contact with the underside of the holder and distal arm, as discussed above. The DC brush motor is an example of any motor **810** that is suitable to adjust vertical position of the holder; a DC brush motor may be advantageous because of its simplicity of control and low cost, as precision control is not needed. An electronic brake **812** prevents backdriving of the lead screw **820** and inadvertent vertical movement; however, the brake can optionally be tuned to allow backdriving at a specific load. An absolute position sensor **814** measures the vertical location of the holder, as the holder must be in a center vertical position for the link members to fold without contacting each other. A Z-axis adjustment switch **818**, positioned anywhere on the arm, but advantageously near the distal end of the arm, e.g., at the end of the third link member, may be a Hall-effect switch to activate motor **810**. Activation of motor **810** causes the motor **810** to rotate the lead screw **820** in its lead nut (not illustrated), thus causing the four-bar linkage of the third link member to change shape and thereby raise the third link member into the angled orientation described herein, e.g., with reference to FIG. **7**, and thus raise the medical device holder **20** upwardly.

[0133] Arm **30** may, in some embodiments, also include one or more additional components which may inhibit or prevent patient harm due to contact with the underside of the holder, including: a counterbalance for a flip-up arm, torque sensors, kick panels/switch array, proximity sensors, and counterbalance wrist and tray in Z-axis parallelogram. Additionally, arm **30** may include a hard

stop and a pivot point counterbalance. More specifically, if provided, a torque sensor could measure patient load, output of which may be used to autonomously drive the arm out of the way using a motor. A kick panel switch array is positioned to recognize any contact with an object, such as with the patient, any may need to cover every surface of the holder. As with the torque sensor, its output may be used to autonomously drive the arm out of the way using a motor. Proximity sensors can be implemented to detect when was an object is near the sensor and thus their outputs may be used to set off an indicator (e.g., an alarm), move the arm, or both, and may have to cover all surfaces of the holder and have to 'see' through the drape. A counterbalanced wrist and tray in a Z parallelogram involves providing a counterbalance within the most distal (third) link member so that it would act like the floating hand, which may warrant precise friction control of backdriving the lead screw.

[0134] In some embodiments, medical device holder **20** may be attached to and stored with arm **30** in the retracted configuration. In such a configuration, the medical device holder **20** may be intended to be reused and provided with a sterile drape, rather than sterilized for each use. For example, FIGS. **10B-1** and **10B-2** illustrate an embodiment of a medical device holder **1020**, which may be used as the medical device holder **20**, coupled to an embodiment of an arm **1030**, which may be used as the arm **30**. As shown in FIG. **10B-1** and **10B-2**, a medical device holder **1020** can be coupled to arm **1030** along a linear guide rail system **1050** to allow the medical device holder **1020** to be slid back over the arm **1030** such that the arm **1030**, along with the medical device holder **1020**, can have a compact, retracted configuration suitable for storage, such as within a base **10** as has been described above with respect to the arm **30**. In some embodiments, a linear guide rail system **1050** may include a guide rail with two shafts mounted to the bottom of the medical device holder **1020** along its entire length. A bearing block (not visible in FIGS. **10B-1** and **10B-2**), is attached to the arm **1030**, and the medical device holder **1020** slides linearly on the shafts with respect to the fixed bearing block. In the embodiment of FIGS. **10B-1** and **10B-2**, the medical device holder **1020** has the configuration as in FIG. **2A-2** and is shown in the folded configuration.

[0135] As shown in FIGS. **10B-1** and **10B-2**, linear rail system **1050** couples the medical device holder **1020** to arm **1030**. Thus, the medical device holder **1020** can slide along rail system **1050** from a first distal position to a second proximal position, such that the second proximal position places the medical device holder over a length of the arm **1030** in the folded configuration. In one embodiment, a handle (not shown) can be provided to manually push or pull medical device holder **1020** relative to arm **1030**.

[0136] With reference to FIG. **10B-3**, another embodiment is illustrated. A medical device holder **20**, shown in broken lines, may include a z-axis control lever **1060** to activate position sensor switches **1062** located at the distal rotary joint of the arm via pushrods **1064**. The pushrods **1064**, located under the medical device holder **20**, are connected to a rotatable handle **1066**. Rotation of the handle **1066** moves the pushrods proximally and distally, causing proximal portions of the pushrods to actuate switches **1062**, the outputs of which may be used to activate a motor, such as motor **810**. Inclusion of z-axis control lever **1060** may eliminate the need for any electrical control wires on the medical device holder **20** and the subsequent need for a rolling wire loop which may take up extra volume. Alternatively, a direct connector may be provided between a lever and the sensor(s), with a rolling loop for the wires using a segmented energy chain to control the flexing of the wires.

[0137] In yet additional embodiments, a mechanism is included which ensures that the medical device holder **20** is fully extended upon arm deployment. This may assist to both prevent collision with the third link member when it is moved vertically by the motor, and also to allow rotational movement of the medical device holder without damaging an installed drape. Such a mechanism may be positioned just above the linear rail (e.g., **210**) in the base (e.g., **510**). The medical device holder **20** is raised and lowered with respect to the most distal rotary arm joint, depending on the deployment position. When retracted, the medical device holder **20** is lifted up using the floating

hand parallelogram linkage by contact with a sliding cam surface fixed to the base. Starting from the fully retracted position within the base (reference being made to FIGS. **6A-6C**), the following sequence occurs. The first link member is locked into the base by the detent, as described above. Pulling on the distal end of the medical device holder results in the holder sliding on its linear slide until it is fully extended, at which time the cam surface on the base allows the medical device holder to move down and drop into a recess formed in the most distal rotary link member, thereby locking the holder in a fully extended position. This action trips a lever attached to the base that releases the first link member and allows the entire arm to slide out of the base. Once the arm is out of the base, continued pulling on the medical device holder results in unfolding of the second link, as described above. After a small amount of unfolding, the lever that is contained within the first link member (also described above) becomes locked into the base, preventing the first link member from retracting into the base. Continued unfolding then can occur to prepare the arm for draping, including unfolding the trough bar of the holder.

[0138] Retracting the arm is the reverse of the foregoing, once the motorized parallelogram is made level. This may be enforced by user-messaging based on the z-axis parallelogram position sensors and the drape presence sensor.

[0139] Arms **30** in accordance with various exemplary embodiments also may have various transmission line routing features disposed along the arm or attachable thereto so that such transmission lines can be routed along an entire length of the arm in an organized manner from surgical instruments on the medical device **20** holder to auxiliary function supply equipment. For example, the various hooks, such as the hook described with reference to FIG. **4**, cable guides, clips, and other mechanisms that can support and route transmission lines can be attached to the arm. Moreover, as shown in the embodiment of FIGS. **11A** and **11E**, described further below, an arm **30** in accordance with the present disclosure can have receiving features that allow sterile transmission line routing structures to be removably received within the arm, such as at the locations of the joints where link members of the arms are coupled together. The arm can directly receive such members, or they can be provided as part of sterile draping that is intended to be placed over the arm, which will be described in more detail below. In an exemplary embodiment, each link member is approximately 2.75 inches wide×3 inches tall. First link member may be 10 inches long, second link member may be 9.5 inches long, and third link member may be 14 inches long, with a joint diameter of 2.75 inches. Link lengths may be optimized so that proper extension into the sterile field can be obtained, and that the arm may be folded into the base cavity without contacting the side of the cavity and without protruding excessively out of the base.

[0140] FIGS. **17-26** illustrates yet further embodiments of various mechanisms that can be included in or used in conjunction with embodiments of the arm **30**. In FIGS. **17-26**, an embodiment of the arm **30** that has three link members is illustrated, and various parts of the arm **30** are referred to using references numbers that were used above in relation to similar parts of the arms **530** and **630**. FIG. **17** shows an exemplary embodiment of a locking mechanism **1700**. In general terms, a locking mechanism, as described herein, is provided for staged deployment of an arm **30**. More specifically, a locking mechanism **1700** controls the motion of the link members of the arm **30** relative to each other, and to the base, from a folder, stowed configuration when pulling on an end of the medical device holder **20**. According to one embodiment, the medical device holder **20** may first slide out, then the first link member **560**, **660** is allowed to deploy by sliding on rail **210**, **610**. Then the links are permitted to pivot and deploy until in a fully extended configuration, such as that illustrated in FIG. **17**.

[0141] With continued reference to FIG. **17**, an arm **30** includes a first link member **560**, **660**, a second link member **563**, **663**, and a third link member **565**, **665**. A medical device holder **20** is attached to the distal end of the arm **30**, and more specifically to the distal end of the third link member **565**, **665**. FIG. **17** also illustrates an exemplary embodiment of a breakaway mechanism **1702**, as well as the rail **210**, **610**. Breakaway mechanism **1702** is described in greater detail below.

[0142] FIG. 18 illustrates a sectional view, with portions broken away, of a first link member **560, 660**, joined with a second link member **563, 663**, at a rotatable joint **573**. With continued reference to FIG. 18, an exemplary embodiment of the breakaway mechanism **1702** is illustrated. The breakaway mechanism **1702** includes a structural yoke **1706** which at least partially surrounds the proximal end portion **1710** of the first link member **560, 660**, and which is mounted to the bearing block **567**. The yoke **1706** may be attached to the proximal end portion **1710** at a pivot pin **1712** so that the yoke and proximal end portion **1710** can pivot relative to each other. A recess **1714** is formed on an inner surface of the proximal end portion **1710** and a vertically aligned recess **1716** is formed on the oppositely facing inner surface of the yoke **1706**. A spring **1718**, which may be a coil spring, a set of stacked disc or wave springs, or the like, has a preselected spring force, for reasons described elsewhere herein. The spring **1718** is positioned between the recess **1714** and the recess **1716**, by which spring preloads the entire arm **30** up against a hard stop **1708**. As the yoke **1706** is biased against the proximal end portion **1710**, both can pivot relative to each other about pivot pin **1712** against the force of the spring **1718**. The hard stop **1708** is attached to or formed with upper portions of the yoke **1706**, so that the range of rotary motion of the first link member **560, 660** relative to the yoke **1706** is limited in one direction. FIG. 18 also illustrates the bearing block **567** interacting with the rail **210, 610** as described elsewhere herein.

[0143] Bearing block **567** may include one or more notches or recesses **1736** on a top surface thereof, for reasons that will be described in greater detail elsewhere here in. Hard stop **1708** may take a number of different forms, with the exemplary embodiment of FIG. 18 being a plate attached to the yoke **1706** adjacent to the upper surface of the link **560, 660**. In this manner, if the arm **30** is overloaded, e.g., is leaned on by a practitioner, the arm **30** can deflect downwards against the force of the spring **1718**, as the proximal end portion **1710** of the link **560, 660**, rotates about the pivot pin **1712** relative to the yoke **1706**. When the overload condition is removed, the force of the spring **1718** between the proximal end portion **1710** and the yoke **1706** forces the arm **30** to resume the configuration illustrated in FIG. 18 until it reaches hard stop **1708**. Thus, a breakaway mechanism as described herein may inhibit or prevent overbalance due to vertical overloading of medical device holder **20** or arm **30**. Optionally, a position sensor (not illustrated) may be mounted on the hard stop **1708**, e.g., at the tip of hard stop, that detects when the hard stop is no longer in contact with the first line (i.e., an overload condition occurs), and thus the system can warn the user about potential overload/overbalance.

[0144] Turning now to FIG. 19, an exemplary locking and deployment mechanism **1700** (also referred to herein as locking mechanism) is illustrated. The exemplary locking mechanism **1700** may include a first generally horizontally extending finger **1722** and a second generally horizontally extending finger **1724**, which fingers are joined together at one or more pivots **1732**. A spring may be employed at pivots **1732** to bias the first and second fingers **1722, 1724** into an unlocked or locked orientation, as will be described in greater detail elsewhere herein. The locking mechanism **1700** further includes a pivot pin **1726** about which a lock arm **1728**, which extends generally longitudinally, is pivotally secured to a linkage support **1720** which in turn may be attached to the base **10**. The lock arm **1728** includes at least one detent **1730** at or adjacent to a proximal end of the lock arm **1728**, as illustrated in FIG. 19. According to further embodiments, the one or more springs which bias the lock arm **1728** and the fingers **1722, 1724** into a locked or unlocked configuration may be located at other positions in the locking mechanism **1700**, such as acting directly between the lock arm **1728** and the support **1720**, as will be readily apparent to those of ordinary skill in the art. FIG. 19 also illustrates a pivoting ramp **1704** which, when push by the medical device holder **20** as will be described elsewhere herein, pivots to engage distal portions of finger **1722** and thereby engage the locking mechanism **1700**. A fixed ramp **1768** is fixedly positioned relative to the linkage support **1720**, against which the pivoting ramp **1704** is hingedly attached so that it can pivot between the configuration of FIG. 19 and the configuration of FIG. 20, described in greater detail below. FIG. 19 also illustrates first link member **560, 660**, second link

member **563, 663**, yoke **1706**, end portion **1710** of the first link member, as well as rail **210, 610**.

[0145] With the reference to FIG. **20**, an exemplary arm **30** and medical device holder **20** are illustrated, with the arm in a folded configuration as similar to configurations described elsewhere herein. The arm **30** may include first link member **560, 660**, second link member **563, 663**, and third link member **565, 665**. In the illustration of FIG. **20**, the medical device holder **20** has been pushed back, that is, proximally, toward the yoke **1706**, which causes a bottom surface **1734** of the medical device holder **20** to push against the pivoting ramp **1704** and push the pivoting ramp **1704** down. When the pivoting ramp **1704** is pushed down by the medical device holder **20**, it rotates down relative to the fixed ramp **1768** and portions of the pivoting ramp **1704** push down against the distal portions of the first finger **1722**. When the first finger **1722** is pushed down, because it is pivotally connected to the second finger **1724**, the second finger **1724** pivots upwardly, allowing the lock arm **1728** to pivot about pivot **1726**. When lock arm **1728** pivots upwardly, its proximal end, and more particularly the detent **1730**, pivots downwardly and engages against the top surface of the bearing block **567**, and more particularly into the notch or recess **1736**. In this manner proximal motion of the medical device holder **20** relative to the arm **30** causes the mechanism to engage and inhibit or prevent first link member **560, 660**, and therefor the entire arm **30**, from further motion, which in turn permits medical device holder **20** to be further pushed proximally, as described in greater detail below.

[0146] FIG. **21** illustrates an exemplary embodiment arm **30** which includes an internal lock mechanism for the first link member **560, 660**. The mechanism includes a striker block **1738** which is mounted to the base **10** and includes a recess therein oriented toward the first link member. A pivoting lock link **1740**, which may be J- or L- or T-shaped, is mounted within the first link member **560, 660**, and is actuatable between locked and unlocked positions, as described in greater detail below.

[0147] With continued reference to FIG. **22**, internal portions of first link member **560, 660**, are illustrated. The locking link **1740** includes a latterly extending locking finger **1742**, which extends through the body of the link member, and into the recess in the striker block **1738**, thereby locking the link member **560, 660** from movement relative to the striker block **1738** and, therefore, relative to the base **10**. The locking link **1740** may include a longitudinally extending arm **1744** which may be connected at a pivot pin **1746** to the link member **560, 660**. Extending further distally from the pivot pin **1746**, the locking link **1740** includes a distal portion **1747** which extends generally longitudinally toward the second link **563, 663**. The distal portion **1747** is biased in one direction by a spring **1750** which may be housed in a recess in a block **1752**, and the distal portion **1747** may include a detent **1754** to engage an end of the spring **1750** and thus retain the spring against the distal portion. A distal tip **1748** of the distal portion **1747** is located adjacent to a block, tab, or protuberance **1756** which is attached to or otherwise rotates with second link **563, 663**.

[0148] With reference to FIG. **23**, the arm **30** is shown in a folded configuration, as described elsewhere herein. As the second link **563, 663** rotates relative to the first link member **560, 660**, the block **1756** moves relative to the first link member **560, 660**, and engages against the distal tip **1748** of the locking link **1740**. This causes the locking link **1740** to rotate about pivot **1746**, causing the laterally extending locking finger **1742** to move out of the recess in the striker block **1738**, and thus unlock the link member **560, 660** and allow it to slide longitudinally along the rail **210, 610**.

[0149] FIG. **24** illustrates yet further embodiments. The medical device holder **20** may include rails **1050** which may be received in rail blocks **1758** mounted to the connecting member **693**. As described elsewhere herein, the connecting member **693** includes pivots (i.e., via needle bearings **690**) which permit the connecting member **693** to tilt upward. A block **1764** is attached to the connecting member **693** and may include a recess **1762** in which a finger **1760** extends and prevents the tray from moving proximally or distally relative to the connecting member. The finger **1760** is fixedly attached to the third link member **565, 665**. With reference to FIG. **25**, the medical device holder **20** has moved proximally relative to the third link member **565, 665** (as illustrated in

FIG. 24), and a proximal portion **1766** of the tray has engaged the pivoting ramp **1704** and has pushed the pivoting ramp **1704** to rotate downward relative to the fixed ramp **1768**. The proximal portion **1766** of the medical device holder **20** is caused to move upward as it engages against the fixed ramp **1768**, causing the finger **1760** to no longer be held in the recess **1762**, and thus permits medical device holder **20** to continue to be pushed proximally (that is, to the right in FIG. 25), because the finger **1760** no longer locks the medical device holder **20** at recess **1762**. This proximal motion of the medical device holder **20**, as discussed elsewhere herein, also causes the lock arm **1728** and finger **1724** to pivot and lock the arm **30** to the bearing block **567**.

[0150] FIG. 26 illustrates a reverse angle view of the arm **30** in a folded configuration, as compared to the view of FIG. 25. As illustrated in FIG. 26, the medical device holder **20** is in a position to engage against the pivoting ramp **1704** to cause the finger **1724** to rotate upward and to cause the lock arm **1728** to rotate downward, causing detent **1730** to engage against the recess in the bearing block **567**, as described elsewhere herein.

Drapes

[0151] To permit an arm in accordance with various embodiments of the present disclosure to be a reusable structure with sufficient stability, and optionally including sophisticated electronics and controls, extending from a base located in a nonsterile field to a sterile field, sterile draping over at least the arm can be used. Thus, it is envisioned that a sterile drape can have a sleeve-like configuration, which can be provided as an enclosed sleeve with an opening at one end to slide the sleeve over the arm. Alternatively, the sterile drape can be generally elongated and planar and configured to wrap around the arm and then be secured along its length to form the sleeve over the arm. In either case, a sterile drape can be secured over the arm, while having the medical device holder provided as a sterile component.

[0152] In an alternative embodiment, as has been discussed above, a sterile draping that covers both the arm and the medical device holder is contemplated. Such sterile draping would be useful so that medical device holders also can be reused without needing to be sterilized between each procedure and/or without needing to be made sterilizable (e.g., autoclavable). Moreover, a sterile drape in accordance with various embodiments can be used in conjunction with a medical device holder having an open frame member such that when the sterile drape is placed over the medical device holder it can form a well that can hold medical devices in a safe, secure, and convenient manner. Further, a sterile drape, whether extending along the arm only or both the arm and the medical device holder, can be provided with transmission line routing features to assist with routing and organization of transmission lines.

[0153] FIGS. 11A-13B show embodiments of sterile drapes configured to be placed over a three-link jointed arm and a medical device holder having a configuration as discussed with reference to the embodiment of FIG. 2A-1-2A-5. For example, the sterile drapes may be designed to cover the arm/medical device holder combination of FIGS. 10A-1-10B-3. FIGS. 11A-11D illustrate an embodiment of a sterile drape **1200**, which receives the medical device holder **20** and at least a portion of arm **30** and thus provides a sterile environment all around the holder **20** and received portions of the arm **30**. FIGS. 11E and 11F illustrate another embodiment of a sterile drape **1300**, which is placed over the medical device holder **20** and the arm **30** without necessarily receiving the holder **20** and arm **30** within the drape **1300** or covering all sides of the holder **20** or arm **30**.

[0154] FIGS. 11A and 11B show two perspective views of an exemplary embodiment of a sterile drape **1200** in position over an arm **1130** and medical device holder **1120**, which can have a configuration of the embodiments of FIGS. 2A-1-2A-5 and 10A-1-10B-3. FIGS. 11C and 11D show perspective views of the drape **1200** removed from the arm **1130** and medical device holder **1120**. Drape **1200** is configured to provide a sterile environment entirely around medical device holder. Additionally, drape **1200** is configured to provide a sterile environment entirely around the arm from the distal end where it couples to the medical device holder and proximally to a position of at least a distal end portion of the first link member where the arm attaches to a base. This

provides a large extension of the sterile field, thus providing greater workflow to a user during a medical procedure.

[0155] The drape **1200** has a configuration such that a portion **1220** of the sterile drape **1200** is received in the opening **1159** defined by a frame portion **1160** of the medical device holder **1120** to form a well/trough configured to receive and hold medical devices therein. As shown in FIG. **11B**, in an exemplary application, the trough formed by placing the portion **1220** of the drape **1200** around the medical device holder **1120** provides a convenient storage area for an imaging instrument, such as an endoscope **1115**. The portion **1225** of the drape **1200** that covers the tray portion **1150** of the medical device holder **1120** can be fitted over the tray portion and include an absorbent material layer **1227** on its outer surface that is placed over the support surface of the tray portion **1150**. The absorbent material layer **1227** can be formed of, for example, cotton, and may be used to absorb any liquids, such as blood and other bodily fluids, that are retained on the medical instruments placed on the tray portion **1150** of the draped medical device holder **1120**. In addition to absorbing fluids from medical devices, the absorbent material layer **1227** also may provide protection from a sharp end of a medical device damaging the sterile drape **1200**. The absorbent material layer **1227** may be adhered to the portion **1225** of the drape using any well-known attachment means such as, for example, tape, adhesive, hook-and-loop-pile type fasteners (e.g., Velcro) or stitching (e.g., sewn together with the portion **1225** of the drape **1200**). While the majority of the supporting surface of the tray portion **1150** of the medical device holder **1120** is covered with the absorbent material layer **1227** of the drape **1200** in the embodiment of FIGS. **11A** and **11B**, one of ordinary skill in the art would appreciate that the absorbent material layer **1227** can be of a variety of sizes and still fall within the scope of the disclosure. For example, the absorbent material layer **1227** may be placed only on one end or around the perimeter of the tray portion **1150** of the medical device holder **1120**, or in any location that is more likely to receive soiled portions of medical devices. In an exemplary embodiment, a drape may include an absorbent material portion approximately 29 inches×7 inches, and a trough portion approximately 30 inches×6 inches×6 inches.

[0156] At least the portion **1220** of the drape **1200** that forms the trough and/or the portion **1225** that is disposed over the tray portion **1150** of the medical device holder can be made of a same or different material than a material of the portions of the drape covering the arm **1130**. In an embodiment, the portions **1220**, **1225** can be made of a thicker, more durable material (e.g., plastic) than other portions of the drape, to provide sufficient strength to hold the height of medical devices placed in the trough and to provide protection additional against puncture and tears. The material(s) of drape, in particular where the drape lies over the medical device holder **1120**, can be so as to allow the drape **1200** to lie relatively flat and unwrinkled, in particular over the tray portion **1150** of the medical device holder **1120**. In some embodiments, one or both portions **1220**, **1225** are formed of a polyurethane (PU) film and the remaining portions of the drape **1200** are formed of low-density polyethylene. Additionally, the material of one or both portions **1220**, **1225** can have a thickness that differs from remaining portions of the drape **1200**. In one embodiment, one or both portions **1220**, **1225** of the drape **1200** has a thickness of about 0.004 inches and other portions of the drape **1200** have a thickness of about 0.0012 inches. When laid flat, an embodiment of a drape **1200** is approximately 18 inches wide×32 inches long, not including any pockets, if provided.

[0157] Drape **1200** can have one or more features to help secure drape **1200** in place and the medical device holder **1120** and/or arm **1130**. For example, as show in FIGS. **11A-11D**, drape **1200** includes cuff tape **1240** to help cloak and wrap drape **1200** over medical device holder **20** and/or arm **30**. Additionally, drape **1200** can include cinch tape **1243** along a length of drape **1200** where it covers the arm **1130**. Once drape **1200** is initially disposed over, for example, arm **1130**, drape **1200** may be bulky in certain areas. Cinch tape **1243** may be used to reduce this bulk so that drape **1200** can be made to more closely adhere to the shape and size of arm **1130**. In various embodiments, drape **1200** also can include side pockets **1245** that are appended to one or more

opposing lateral sides of the portion of the drape **1200** that covers the medical device holder **1120**. These pockets **1245** can provide additional storage, and cuff tape **1240** can also be used to identify the location of pockets **1245**.

[0158] To further assist retention of the drape **1200** in position over the medical device holder **1120**, one or more clips can be used. As shown in FIGS. **11A** and **11B**, clips **1270** are U-shaped plastic members that snap over the drape **1200** to the frame portion **1160** and protruding lip surrounding the tray portion **1150**. The clips **1270** can secure the drape **1200** in place and help to prevent ballooning of the drape **1200** when medical devices are placed in either the trough formed by portion **1220** or the on the tray portion **1150**, thereby enhancing protection of medical devices against falling off the medical device holder **1120**. Clips **1270** can be provided around the outer perimeter of the medical device holder **1120** as well as along a center separating the tray portion **1150** from the frame portion **1160**. Clips **1270**, when positioned in the center of the tray portion **1150**, also prevent the tray portion of the drape **1200** from shifting or sliding when devices are inserted into the trough formed by portion **1220**. The clips **1270** in the center can also be provided with vertically protruding guide ramps **1271** that can assist with guiding transmission lines up and over medical devices that may lie on the tray portion **1150** or in the trough formed by portion **1220**, to provide an additional measure of protection from transmission lines accidentally engaging medical devices and causing unintended movement thereof.

[0159] As shown in FIG. **12B**, a planform schematic view, and a perspective schematic view in FIG. **12A** of the portions **1220** and **1225** of drape **1200** are shown, along with a detailed view of a clip **1270** (without the transmission line guide ramp) in FIG. **12C**. While various exemplary embodiments contemplate the clips **1270** being provided as separate components, in an embodiment, the clips **1270** may be attached to the drape **1200** and come as part of the drape structure, as shown for example, in the embodiment shown in FIGS. **11C** and **11D**, which show perspective views of the drape **1200** removed from the arm **1130** and medical device holder **1120** in FIGS. **11A** and **11B**. Clips **1270** can be formed of, for example, injection molded acrylonitrile butadiene styrene (ABS). Clips **1270** can include protrusions **1273** (only one being shown) on an internal surface of the leg members forming the U-shape to provide a snap-fit retention with the lip surrounding the tray portion and frame member of the medical device holder.

[0160] In another embodiment, an end cap can be fit within the drape to better form the trough part of the drape and to further increase resistance to puncture. Reference is made to FIGS. **13A** and **13B**, which show an end cap **1329** in isolation (FIG. **13B**) and fitted within the trough portion of a drape in position over an end of the frame member of a medical device holder (FIG. **13A**). A similar trough end cap **1329** can optionally be provided at an opposite end of the medical device holder and trough. As shown, trough end cap **1329** can include a relatively thicker plastic or paperboard material, such as, for example, low-density polyethylene (LDPE) or coated paperboard that can be snap-fit over the end of the trough and medical device holder frame. In some embodiments, as shown in FIGS. **13A** and **13B**, trough end caps **1329** may be a separate member that is snap fitted onto medical device holder **1120** over drape **1200**. Trough end caps **1329** may be configured to prevent drape **1200** from being punctured by the tip of a medical instrument when the instrument is disposed (or is in the process of being disposed) within the trough formed by portion **1220** of the drape **1200**, and also provide a smooth surface over which to run various transmission lines (as illustrated in FIG. **13A**).

[0161] Drapes **1200** and **1300** also can include (or be used in conjunction with) cable guides **1280** which comprise transmission line routing features, such as one or more clips and/or guideposts to receive and route transmission lines from medical devices supported on the medical device holder to equipment outside of the sterile field. In an embodiment, the cable guides **1280** also can help secure the drape **1200**, **1300** to the arm **1130**. The cable guides **1280** can be provided at joints **1131** of the arm **1130** and can be permitted to swivel to allow for slack in and prevent pulling of transmission lines as the arm **1130** and/or medical device holder **1120** are moved. Additionally, the

cable guides **1280** can minimize any shifting or moving of drape **1200**, **1300** when items are placed in the trough portion.

[0162] Reference is made to FIGS. **11A-11F** for various views of examples of how the cable guides **1280** can be used in relation to drapes **1200**, **1300** and/or arm **30**. The detailed view of FIG. **11E** shows transmission lines TL routed through the routing features of the cable guides **1280**. As shown in the FIGS. **11A-11F**, cable guides **1280** can be secured along a length of the drape **1200**, **1300** that covers the arm **30** to organize and direct transmission lines. Generally, embodiments of the cable guides **1280**, as shown in FIGS. **11A-11F** and **14A-1-14D**, have a U-shaped central portion formed by guidepost arms, which define an opening to receive and route transmission lines. Some embodiments of the cable guides **1280**, as shown in in FIGS. **11A-11F** and **14D**, also include one or more side arms extending laterally from the guidepost arms to define an additional opening to receive and route an additional transmission line. In other embodiments of the cable guides **1280**, as shown in FIGS. **14A-1-14C**, such side arms are omitted. Specific embodiments of the cable guides **1280** are illustrated in FIGS. **14A-1-14D** and are described in greater detail below.

[0163] FIG. **14D** illustrates an embodiment of a cable guide **1290**, which can be used as the cable guides **1280**. In one embodiment, as shown in FIG. **14D**, the cable guide **1290** comprises guidepost arms **1291** that extend in one direction and side arms **1293** that extend laterally from the guidepost arms **1291** in another direction, on one or both sides. The side arms **1293** comprise clips **1297**, for example at an end of the side arms **1293**, that are intended to receive transmission lines associated with a monopolar or a bipolar electrosurgical instrument. Separating such transmission lines can help to avoid capacitive coupling between the lines if they were located too close together. It may be sufficient to have only one side arm **1293** and route one of the bipolar or monopolar electrical transmission line through the side arm **1293** and the other through the main opening **1298** of the cable guide **1290**.

[0164] As mentioned above, the cable guide **1290** with side arms **1293** can be configured to be attached to the drape **1200** and through the drape **1200**, **1300** to the arm **1130**. FIG. **14D** shows an exemplary structure of a cable guide **1290** with side arms **1293** for attachment to a drape **1200** and for subsequent attachment of the drape **1200**, **1300** to an arm **1130**. The cable guide **1290** includes opposing guidepost arms **1291** that define boundaries of an opening **1298** between the guidepost arms **1291** to receive and route a bundle of transmission lines (not illustrated). Retention elements **1292** extend from top portions of the guidepost arms **1291** and further define boundaries of the opening **1298**. The retention elements **1292** are oriented and slanted toward each other to provide a barrier for transmission lines, which helps to retain transmission lines within the opening **1298** formed between the guidepost arms **1291**. Bottom portions of the guidepost arms **1291** are coupled to a base portion **1295**. In this embodiment, the cable guide **1290** further includes a separate base **1294** that can be disposed on the inside of drape **1200**, **1300**. The upper portion of the cable guide **1290** that is outside the drape **1200** (e.g., base portion **1295**) can fasten through the drape **1200**, **1300** to the base **1294**. Such fastening can occur via a mechanical connection or via magnetic connection. The guidepost arms **1291** can be flexible, e.g., are formed of a flexible material, such as a flexible plastic, e.g., ABS, and may bend in order to accommodate a large number of transmission lines and to allow a user to easily place them within the opening.

[0165] In some embodiments, the side arms **1293** can extend laterally outward from one or both of the guidepost arms **1291**, and can comprise clips **1297** that define openings **1299** that can receive, separate, and route transmission lines, such as, for example bipolar or monopolar electrical energy transmission lines as discussed above.

[0166] FIGS. **14A-1-14C** depict other structural arrangements and features of embodiments of the cable guides **1280** in accordance with the present disclosure. FIGS. **14A-1** and **14A-2** show perspective and top plan views, respectively, of a cable guide **1350** according to yet another embodiment, which can be used as the cable guides **1280**. With reference to FIG. **14A-1**, a cable guide **1350** includes a pair of upwardly extending guidepost arms **1352** and slanted retention

elements **1392** extending from the guidepost arms **1352**, which together define at least some boundaries of an opening **1356** between them. The pair of upwardly extending guidepost arms **1352** are integral with the retention elements **1392** and extend up from a pair of latterly extending members **1354**. Together, the retention elements **1392**, the guidepost arms **1352**, and the members **1354** define the opening **1356** to accommodate transmission lines. The members **1354** extend laterally from a cylindrical or semi-cylindrical boss **1358** (also referred to herein as a base portion) which has an optionally hollow interior. One or more ramped, snap fit features **1360** are formed on an exterior surface of the boss **1358**. The boss **1358** is sized to cooperate with a receptacle **1364** formed in a portion **1362** of the arm **30** and/or the holder **20**. The receptacle **1364** defines an opening or hole **1365** into which the boss **1358** may be received. The portion **1362** includes a ramp **1366** which is sized to cooperate with the snap fit feature **1360** on the boss **1358** so that the boss **1358** can be forced into the receptacle **1364** and held therein by the snap fit features **1360** and ramps **1366**, while still permitting the boss **1358** to swivel within the receptacle **1364** (and hence permitting the cable guide **1350** to swivel relative to the arm **30**). While the boss **1358** is being forced into the receptacle **1364**, the snap fit feature **1360** and/or ramps **1366** flex as the snap fit feature **1360** passes over the ramps **1366**, and then once the snap fit feature **1360** has passed the ramps **1366**, the snap fit feature **1360** and/or ramps **1366** snap back into place to hold the boss **1358** within the receptacle **1364**.

[0167] FIGS. **14B-1** to **14B-3** illustrate another exemplary embodiment of a cable guide **1380**, which may be used as the cable guide **1280**. The cable guide **1380** is similar in some respects to cable guide **1350** and common reference numerals have been used to designate similar or the same features. Cable guide **1380** includes a rib structure **1370** on the interior surface of one or both of the guidepost arms **1352**. The rib structure **1370** helps to provide rigidity to the guidepost arms **1352** and to inhibit transmission lines from snagging on the arms **1352**. In addition, a snap fit feature **1372** is provided on an exterior surface of boss **1358**. The snap fit feature **1372** may also be formed as a ribbed structure similar to that of rib structure **1370**. Similarly, snap fit feature **1372** is more rigid than the snap fit feature **1360** of cable guide **1350**, but it performs the same general function of permitting the cable guide **1380** to be snapped into a receptacle in a portion of the arm **30**, such as portion **1362**, and permit the cable guide **1380** to swivel relative to that portion.

[0168] FIG. **14-C** illustrates another exemplary embodiment of attaching cable guide **1280** as described herein to a drape **1200**, **1300**. As illustrated in FIG. **14C**, a hole **1384**, which may be formed by three slits arranged in the shape of an H, is made in the drape **1200**, **1300** at a location that is (or will be) positioned over a receptacle **1386**. The receptacle **1364** in the portion **1362** of the arm **30** or holder **20** may be used as the receptacle **1386**. The drape **1200**, **1300** comprises two opposing flaps **1385** at the hole **1384**, which may be formed as a result of making the slits. The opposing flaps **1285** at least partially bound and define the hole **1384**. A cable guide **1350**, **1380** is then inserted through the hole **1384** into the receptacle **1386**, displacing the opposing flaps **1385** into the receptacle **1386** as a result. A pair of fingers **1382** (also referred to herein as hole-closing members) adjacent to the receptacle **1385** are then pushed towards the receptacle **1386**, which raises the flaps **1385** to be coplanar with the rest of the material of the drape **1200** and thus to at least partially cover the boss **1358**. Lastly, a piece of tape **1387** or the like (also referred to herein as a sealing member) is then placed into the interior of the boss **1358** of the cable guide **1350**, **1380** from above and secured to the drape **1200**, **1300**, thus capturing the cable guide **1350**, **1358** in place and sealing the hole **1384**.

[0169] The cable guides **1280** and other transmission line routing features described herein may be integral components with a drape (e.g., drape **40** or **1200**), and they may be packaged together with a drape. Alternatively, they can be provided as separate components and attached to the drape when ready for use.

[0170] FIGS. **11E** and **11F** provides another perspective view of a drape **1300** having many of the features as drape **1200** described above, placed over a medical device holder and arm in accordance

with an exemplary embodiment of the present disclosure. As shown in FIGS. **11E** and **11F**, the drape **1300** may be used in conjunction with cable guides **1280**, clips **1270**, and/or cinch tape **1243**, as described above. As shown in FIG. **11F**, the drape **1300** may also have a portion **1220** forming a trough and/or side pockets **1245**, as described above.

[0171] In some embodiments, it is contemplated that separate drapes are provided over the medical device holder and the arm respectively. Thus, the size and shape of a first drape may be tailored for medical device holder **1120**, and the size and shape of a second drape may be tailored for arm **1130**. Various other features as described above may still be provided as the respective first and second drapes.

Base

[0172] As discussed above, base **10** may be any of a variety of structures. In an exemplary embodiment, the base **10** is an auxiliary system cart that includes equipment for supplying and controlling various auxiliary surgical functions, such as, for example, image processing, electrocautery energy generation, system central processing, and/or insufflation/suction. In other embodiments, base **10** may be any cart or fixed base sufficient to provide a stable support structure for an arm and medical device holder in accordance with various exemplary embodiments described herein. It is also contemplated that base **10** may be a patient's operating table or another telesurgical system component, such as a manipulator system or a master control unit (e.g., surgeon console). By way of a non-limiting example, one such base is 26 inches wide×27 inches deep×60 inches high.

[0173] As shown in FIGS. **15A** and **15B**, the base to which an arm **1530** and medical device holder **1520** attach can be a wheeled auxiliary system cart **1510** (also referred to as base **1510**) that holds one or more auxiliary function equipment **1501**, **1502** (two such equipment components being depicted) and provides control functionality of a telesurgical system. Examples of such auxiliary function equipment include, but are not limited to endoscopic illumination and control units, electrosurgical energy generators, insufflation devices, robot system control and processing equipment, specialty instruments control units, saline and fluid dispensing, and suction units. The auxiliary system cart **1510** may be used as the base **10**, the arm **1530** may be one of the embodiments of the arm **30** described herein, and the medical device holder **1520** may be one of the embodiments of the medical device holder **20** described herein. In an exemplary embodiment, the arm **1530**, alone or with the medical device holder **1520** attached, can fold and retract into a cavity **1560** provided in the auxiliary system cart **1510**, in a manner similar to that described above with regard to the exemplary embodiments of FIGS. **6A-6E**. In an embodiment, the cavity **1560** may be Approximately 8 inches tall×9 inches wide×24 inches deep. Although not shown, connector ports, to receive the connectors on the transmission lines of medical devices, can be part of the auxiliary function equipment **1501**, **1502** in order to connect medical devices on the medical device holder to the auxiliary function equipment **1501**, **1502**.

[0174] The base **1510** also includes one or more monitors and/or screens **1540** in order to show various functions and parameters associated with the patient, for example, oxygen concentration and heart rate, and also to show images of a remote surgical site being captured by an image capture device during the surgical procedure.

[0175] Base **1510** may remain outside of the sterile field. But, providing draping on the arm **1530** and the medical device holder **1520**, or otherwise, provides a sterile medical device holder **1520** that allows medical devices supported by the medical device holder to be safely connected to the auxiliary equipment of the base **1510** without compromising the sterile field.

[0176] FIGS. **15A** and **15B** show medical device holder **1520** and arm **1530** in the extended configuration such that these members are disposed outward of cavity **1560**. FIG. **15C** shows medical device holder **1520** and arm **1530** in the retracted configuration such that these members are disposed within cavity **1560**. As shown in FIG. **15D**, cavity **1560** may be enclosed with a door **1565** to protect medical device holder **1520** and arm **1530** when they are stored within cavity **1560**.

The door **1565** may be manually opened and closed, or may be automated. Base **1510** is wide and heavy enough to prevent instability or overbalance due to typical loads applied to the arm or holder. Energy for the medical devices may be provided via electrically conductive cables (not illustrated) from any of the auxiliary or integrated devices described herein.

[0177] FIGS. **16A-16B** show perspective views of exemplary embodiments of a tray and an arm used with a stand. Any of the embodiments described herein may be used with a conventional Mayo stand-type base, as generally illustrated in FIGS. **16A-B**. FIG. **16A** illustrates an exemplary medical device holder **20** used with, or mounted to, a stand **1600**. FIG. **16B** illustrates an exemplary arm **30** used with, or mounted to, a stand **1602**.

Overall System

[0178] In various exemplary embodiments, an arm that provides a mechanical supporting link between a base and a medical device holder, such as, for example, a tray, receptacle, surface, or any combination thereof, may provide a stable structure that facilitates workflow by allowing sterilized medical devices to be connected, routed, and set up prior to and during surgery. The various exemplary embodiments may be used in any combination in a medical procedure in order to provide convenient and easy access to a sterile environment. A user may easily manipulate the location of each of the disclosed components to quickly and efficiently rearrange the layout of the sterile environment. Such may be useful if a user wants to access the surgical site from another angle, or if the user needs to quickly access the surgical site. Additionally, the easy manipulation of the system speeds up the process between pre-op and intra-op. For example, the various exemplary embodiments allow sterilized devices to be easily attached, routed, and staged prior to surgery. The various exemplary embodiments also provide superior cable management and storage compared with conventional systems. Furthermore, as discussed above, the various exemplary embodiments provide for easy and secure storage of an arm and a medical device holder within an outer base.

[0179] As discussed above, the features discussed herein provide instrument and cable storage and organization. Thus, the features discussed herein provide an improved workflow system.

[0180] Further modifications and alternative embodiments will be apparent to those of ordinary skill in the art in view of the disclosure herein. For example, the systems and the methods may include additional components or steps that were omitted from the diagrams and description for clarity of operation. Accordingly, this description is to be construed as illustrative only and is for the purpose of teaching those skilled in the art the general manner of carrying out the present teachings. It is to be understood that the various embodiments shown and described herein are to be taken as exemplary. Elements and materials, and arrangements of those elements and materials, may be substituted for those illustrated and described herein, parts and processes may be reversed, and certain features of the present teachings may be utilized independently, all as would be apparent to one skilled in the art after having the benefit of the description herein. Changes may be made in the elements described herein without departing from the spirit and scope of the present teachings and following claims.

[0181] It is to be understood that the particular examples and embodiments set forth herein are nonlimiting, and modifications to structure, dimensions, materials, and methodologies may be made without departing from the scope of the present teachings. For example, various aspects have been described in the context of an instrument used in a surgical robotic system. But these aspects may be incorporated into hand-held instruments as well.

[0182] Other embodiments in accordance with the present disclosure will be apparent to those skilled in the art from consideration of the specification, and practice of the teachings disclosed herein. It is intended that the specification and examples be considered as exemplary only, with a true scope and spirit being indicated by the following claims.

Claims

1. A medical system, comprising: an arm; a medical device holder coupled or mountable to the arm; a drape configured to cover the medical device holder and at least a portion of the arm in a coupled or mounted state of the medical device holder to the arm; and a cable guide comprising: a base portion coupled or configured to be coupled to a complementary receptacle of the arm; and two opposing guidepost arms extending from the base portion and defining an opening configured to receive and route a transmission line, wherein, in a coupled state of the base portion to the receptacle, the guidepost arms are positioned on a first side of the drape and the arm is positioned on a second side of the drape, the first side being opposite from the second side.
 2. The medical system of claim 1, wherein: the opening is a first opening configured to receive and route a first transmission line along a first path; the cable guide further comprises a side arm extending laterally from one of the guidepost arms and defining a second opening; and the second opening is spaced from the first opening and is configured to receive and route a second transmission line along a second path spaced from the first path.
 3. The medical system of claim 1, wherein: the arm is articulated and comprises a joint; and the cable guide is coupled to the arm at the joint.
 4. The medical system of claim 3, further comprising: a plurality of cable guides including the cable guide; wherein the arm comprises a plurality of joints including the joint, and the plurality of cable guides are coupled to the arm at least at each of the plurality of joints.
 5. The medical system of claim 1, wherein: the medical device holder comprises a frame defining an opening, a portion of the drape is configured to be received within the opening defined by the frame to form a trough configured to receive a medical device.
 6. The medical system of claim 5, wherein: the medical device holder further comprises a tray; and the frame is rotatably coupled to the tray.
 7. The medical system of claim 1, further comprising: a base comprising a cavity, wherein the arm is coupled to the base and moveable between a retracted configuration, in which the arm is received within the cavity, and an extended configuration.
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